

shift ρ , we begin by taking QR decompositions

$$B^T B - \rho I = \hat{Q} R \quad \text{and} \quad B B^T - \rho I = \hat{P} S, \quad (5.9.28)$$

where $\hat{Q}$ and $\hat{P}$ are orthogonal and R and S are upper triangular. If we now define $\hat{B}$ by

$$\hat{B} = \hat{P}^T B \hat{Q}, \quad (5.9.29)$$

we see immediately from (5.9.28) and (5.9.29) that

$$\hat{B}^T \hat{B} = \hat{Q}^T B^T B \hat{Q} = R \hat{Q} + \rho I \quad (5.9.30)$$

and

$$\hat{B} \hat{B}^T = \hat{P}^T B B^T \hat{P} = S \hat{P} + \rho I. \quad (5.9.31)$$

This means that the transformations $B B^T \rightarrow \hat{B} \hat{B}^T$ and $B^T B \rightarrow \hat{B}^T \hat{B}$ are both shifted QR steps. Thus the transformation (5.9.29) effects the QR steps implicitly.

Our objective now is to perform the transformation (5.9.29), and for this we need the matrices $\hat{P}$ and $\hat{Q}$. These come from the QR decompositions (5.9.28), for which we would appear to need $B^T B - \rho I$ and $B B^T - \rho I$. But we want to avoid this; we want to generate $\hat{P}$ and $\hat{Q}$ without performing the QR decompositions explicitly or even building $B^T B$ or $B B^T$.

Each of $\hat{P}$ and $\hat{Q}$ is a product of rotators:

$$\hat{P} = P_1 \cdots P_{m-1} \quad \text{and} \quad \hat{Q} = Q_1 \cdots Q_{m-1}. \quad (5.9.32)$$

If we have these rotators, we can apply them to B , one after the other, to transform B to $\hat{B}$. The challenge is to find an inexpensive way to generate each rotator as it is needed, so that we can effect (5.9.29) efficiently. In Section 5.7 we saw how to do this for the standard eigenvalue problem. A similar development could be pursued here. Instead we will take the shorter route: We will simply describe the implicit QR algorithm and then use the implicit- Q theorem to justify it.

First we will make some observations on the form of $\hat{B}$. If ρ is not an eigenvalue of $B B^T$ and $B^T B$, then the upper-triangular matrices S and R in (5.9.28) are both nonsingular. They can even be normalized so that their main-diagonal entries are positive. The following exercise shows that in this case, $\hat{B}$ inherits the proper bidiagonal form of B .

Exercise 5.9.33 Suppose R and S in (5.9.28) are both nonsingular.

(a) Show that

$$(B B^T - \rho I) B = B (B^T B - \rho I) \quad \text{and} \quad (B^T B - \rho I) B^T = B^T (B B^T - \rho I).$$

(b) Using (5.9.28), (5.9.29), and the result from part (a), show that $\hat{B} = S B R^{-1}$ and $\hat{B}^T = R B^T S^{-1}$.

(c) Show that if B is upper triangular, then so is $\hat{B}$.

- (d) Show that if B^T is upper Hessenberg, then so is $\hat{B}^T$.
- (e) Show that if B is bidiagonal, then so is $\hat{B}$.
- (f) Show that if B is properly bidiagonal, then so is $\hat{B}$.
- (g) Show that if diagonal and superdiagonal entries of B are all positive, then the same is true of $\hat{B}$, assuming that R and S are normalized so that their main-diagonal entries are positive.

□

The results of this exercise all hold and, in particular, $\hat{B}$ is properly bidiagonal, if ρ is not an eigenvalue of $B^T B$ and BB^T . Even if ρ is an eigenvalue, the conclusions still hold for the most part. The only difference is that $\hat{B}$ has a zero in its $(m-1, m)$ position; that is, $\hat{\gamma}_{m-1} = 0$. A careful discussion of this case requires some extra effort, and we will omit it.

Now we can describe the implicit QR step. To get started, all we need is the first column of $\hat{Q}$. From the equation $B^T B - \rho I = \hat{Q} R$ of (5.9.28) and the fact that R is upper triangular, we see that the first column of $\hat{Q}$ is proportional to the first column of $B^T B - \rho I$, which is

$$\begin{bmatrix} \beta_1^2 - \rho \\ \gamma_1 \beta_1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}. \quad (5.9.34)$$

We can easily compute these two numbers; there is no need to compute the whole matrix $B^T B - \rho I$. Let V_1 be a rotator (or reflector) in the $(1, 2)$ plane whose first column is proportional to (5.9.34). Multiply B by V_1 on the right. The operation $B \rightarrow BV_1$ alters only the first two columns of B and creates a new nonzero entry (a “bulge”) in the $(2, 1)$ position. (Draw a picture!)

Our objective now is to return the matrix to bidiagonal form by chasing the bulge. As a first step in this direction, find a rotator U_1 in the $(1, 2)$ plane such that $U_1^T BV_1$ has a zero in the $(2, 1)$ position. This operation acts on rows 1 and 2 and creates a new bulge in the $(1, 3)$ position. Now let V_2 be a rotator acting on columns 2 and 3 such that $U_1^T BV_1 V_2$ has a zero in the $(1, 3)$ position. This creates a bulge in the $(3, 2)$ position. Applying additional rotators $U_2^T, V_3, U_3^T, \dots$, we chase the bulge through the positions $(2, 4), (4, 3), (3, 5), (5, 4), \dots, (m, m-1)$, and finally off of the matrix completely. This completes the implicit QR step. The result is a bidiagonal matrix

$$\tilde{B} = U_{m-1}^T \cdots U_2^T U_1^T BV_1 V_2 \cdots V_{m-1}. \quad (5.9.35)$$

By appropriate choice of the rotators, we can ensure that all of the entries of $\tilde{B}$ are nonnegative.

Exercise 5.9.36 (a) Show that each of the rotators $V_1, V_2, \dots, V_{m-1}, U_1, U_2, \dots, U_{m-1}$ in (5.9.35) is a nontrivial rotator (i.e. rotates through an angle other than 0 or 180

degrees), and at the same time deduce that the entries of $\hat{B}$ satisfy $\hat{\beta}_i \neq 0, i = 1, \dots, m-1$ and $\hat{\gamma}_i \neq 0, i = 1, \dots, m-2$. (It can happen that $\hat{\gamma}_{m-1} = 0$ or $\hat{\beta}_m = 0$. In fact $\hat{\gamma}_{m-1} = 0$ when and only when ρ is an eigenvalue of $B^T B$.) (b) Show that the rotators can be chosen so that all of the $\hat{\beta}_i$ and $\hat{\gamma}_i$ are nonnegative. (c) Show that the off-diagonal entries of $\hat{B}^T \hat{B}$ and $\hat{B} \hat{B}^T$ are all nonzero, except possibly the $(m, m-1)$ entry. $\square$

Letting

$$\tilde{P} = U_1 U_2 \cdots U_{m-1} \quad \text{and} \quad \tilde{Q} = V_1 V_2 \cdots V_{m-1}, \quad (5.9.37)$$

we can rewrite (5.9.35) as

$$\tilde{B} = \tilde{P}^T B \tilde{Q}. \quad (5.9.38)$$

Now we must show that $\tilde{B}$ is (essentially) the same as the matrix $\hat{B}$ produced by (5.9.29).⁸ The desired result is a consequence of the following theorem of “implicit- Q ” type.

Theorem 5.9.39 *Let B be nonsingular. Suppose $\tilde{B}$ is bidiagonal, $\hat{B}$ is properly bidiagonal, $\hat{P}, \hat{Q}, \tilde{P}, \tilde{Q}$ are orthogonal,*

$$\hat{B} = \hat{P}^T B \hat{Q} \quad (5.9.40)$$

and

$$\tilde{B} = \tilde{P}^T B \tilde{Q}. \quad (5.9.41)$$

Suppose further that $\hat{Q}$ and $\tilde{Q}$ have essentially the same first column; that is, $\hat{Q}e_1 = \tilde{Q}e_1 d_1$, where $d_1 = \pm 1$. Then there exist orthogonal diagonal matrices D and E such that $\hat{Q} = \tilde{Q}D$, $\hat{P} = \tilde{P}E$, and

$$\hat{B} = E \tilde{B} D.$$

In other words, $\hat{B}$ and $\tilde{B}$ are essentially the same.

Proof. We note first that $\hat{B}^T \hat{B} = \hat{Q}^T (B^T B) \hat{Q}$ and $\tilde{B}^T \tilde{B} = \tilde{Q}^T (B^T B) \tilde{Q}$. By Exercise 5.9.26, $\hat{B}^T \hat{B}$ and $\tilde{B}^T \tilde{B}$ are tridiagonal matrices, and the latter is properly tridiagonal. Applying the relaxed version of the implicit- Q theorem (Theorem 5.7.24) with $B^T B$, $\hat{B}^T \hat{B}$, and $\tilde{B}^T \tilde{B}$ playing the roles of A , $\hat{A}$, and $\tilde{A}$, respectively, we find that there exists an orthogonal diagonal matrix D such that $\hat{Q} = \tilde{Q}D$. (That is, $\hat{Q}$ and $\tilde{Q}$ are essentially the same.) Using this equation, we can rewrite (5.9.40) and (5.9.41) to obtain

$$\hat{P} \hat{B} = B \hat{Q} = B \tilde{Q} D = \tilde{P} (\tilde{B} D).$$

Defining $C = \hat{P} \hat{B} = \tilde{P} (\tilde{B} D)$, we note that $\hat{P} \hat{B}$ and $\tilde{P} (\tilde{B} D)$ are both QR decompositions of C . Therefore, by Exercise 3.2.47, $\hat{P}$ and $\tilde{P}$ are essentially equal; that is,

⁸We could even prove more; we could show that the rotators $U_1, \dots, U_{m-1}$ and $V_1, \dots, V_{m-1}$ are (essentially) the same as $P_1, \dots, P_{m-1}$ and $Q_1, \dots, Q_{m-1}$, respectively, but we will not take the time to do this.

there is an orthogonal diagonal matrix E such that $\hat{P} = \tilde{P}E$. It follows immediately that $\hat{B} = E\tilde{B}D$. $\square$

Exercise 5.9.42 Show that if $\hat{B}$ and $\tilde{B}$ have $\hat{\beta}_i > 0$, $\hat{\gamma}_i > 0$, $\tilde{\beta}_i > 0$, and $\tilde{\gamma}_i > 0$, for all i , and $\hat{Q}e_1 = \tilde{Q}e_1$ in Theorem 5.9.39, then $\hat{Q} = \tilde{Q}$, $\hat{P} = \tilde{P}$, and $\hat{B} = \tilde{B}$. $\square$

To apply Theorem 5.9.39 to our present situation, note that the first column of the matrix $\hat{Q} = V_1 \cdots V_{m-1}$ of (5.9.37) is the same as the first column of V_1 . This is easy to see if we think of accumulating $\hat{Q}$ by starting with V_1 , then multiplying on the right by $V_2, V_3, \dots, V_{m-1}$, successively. Since each of these is a rotator acting on columns other than column 1, the first column of $\hat{Q}$ must be the same as the first column of V_1 . The latter was chosen to be essentially the same as the first column of the orthogonal matrix $\hat{Q}$ of (5.9.28) and (5.9.29). We can now apply Theorem 5.9.39 to conclude that the matrix $\tilde{B}$ defined by (5.9.29) is essentially the same as the matrix $\hat{B}$ defined by (5.9.35) and (5.9.38). Recalling (5.9.30) and (5.9.31), we see that our bulge-chasing process effects QR iterations on both $B^T B$ and BB^T . This completes the justification of the implicit QR step, at least in the case that the shift is not an eigenvalue.

Exercise 5.9.43 Work out what happens in the case that ρ is an eigenvalue of $B^T B$ and BB^T . You may find the ideas in Exercise 5.6.28 useful. $\square$

Shifts and Deflation

If we set $B \leftarrow \hat{B}$ and perform repeated QR steps with some reasonable choice of shifts, both $B^T B$ and BB^T will tend to diagonal form. The main-diagonal entries will converge to the eigenvalues. If the shifts are chosen well, the $(m, m-1)$ and (m, m) entries of both $B^T B$ and BB^T will converge very rapidly, the former to zero and the latter to an eigenvalue. Of course we do not deal with $B^T B$ or BB^T directly; we deal with B . The rapid convergence of $B^T B$ and BB^T translates into convergence of γ_{m-1} to zero and β_m to a singular value. Once γ_{m-1} becomes negligible (e.g. $\gamma_{m-1} \leq u(\beta_{m-1} + \beta_m)$, where u is the unit roundoff), we can deflate the problem and go to work on the remaining $(m-1) \times (m-1)$ submatrix. After $m-1$ deflations, we are done.

During the whole procedure, all of the γ_k tend slowly toward zero. If at any point one of them becomes negligible, the problem should be reduced to two smaller subproblems, as in (5.9.27).

So far, we have not indicated how to choose the shifts. It is worthwhile to start with a few iterations with $\rho = 0$. If B has any tiny singular values, these will emerge promptly at the bottom, and B can be deflated.⁹ Once the tiny singular values have been removed, we should switch to a shifting strategy that causes the remaining

⁹Recall from Theorem 5.6.21 that a zero eigenvalue of $B^T B$, which corresponds to a zero singular value of B , will emerge in one iteration. Similarly, eigenvalues that are very close to zero will emerge in just a few iterations.

singular values to emerge quickly. To this end, we should try to choose shifts that approximate eigenvalues of $B^T B$ well. For example, it is a simple matter to compute the lower-right-hand 2×2 submatrix of $B^T B$. It is

$$\begin{bmatrix} \gamma_{m-2}^2 + \beta_{m-1}^2 & \gamma_{m-1}\beta_{m-1} \\ \gamma_{m-1}\beta_{m-1} & \gamma_{m-1}^2 + \beta_m^2 \end{bmatrix}. \quad (5.9.44)$$

We are using the symbol B to refer to the current iterate, not the original matrix. Thus m refers to the current dimension of the matrix (some deflations may have occurred), and the β 's and γ 's are the current values, not the original ones. We can compute the two eigenvalues of the matrix (5.9.44) and let the shift ρ for the next iteration be that eigenvalue that is closer to $\gamma_{m-1}^2 + \beta_m^2$. This is the Wilkinson shift on $B^T B$. It is a good choice because it guarantees convergence, and the convergence is rapid in practice.

Exercise 5.9.45 Write a Fortran (or MATLAB or C or ...) program that calculates the singular values of a bidiagonal matrix B . Test it on the matrix $B \in \mathbb{R}^{m \times m}$ given by $\beta_i = 1$ and $\gamma_i = 1$ for all i . Try various values of m . The singular values of B are

$$\sigma_j = 2 \cos \frac{j\pi}{2m+1} \quad j = 1, \dots, m.$$

□

Additional Exercises

Exercise 5.9.46 Let $B \in \mathbb{R}^{m \times m}$ have the form (5.9.24). Since B is $m \times m$, we can say that B has m singular values $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_m \geq 0$. If $\text{rank}(B) = r < m$, the last $m - r$ singular values are zero.

- Show that $\beta_k = 0$ for some k if and only if zero is a singular value of B .
- Suppose $\beta_k = 0$. Show that we can make $\gamma_{k-1} = 0$ by a sequence of $k - 1$ rotators applied to B from the right. The first acts on columns $k - 1$ and k and transforms γ_{k-1} to zero. At the same time it produces a new nonzero entry in position $(k - 2, k)$. The second rotator acts on columns $k - 2$ and k , annihilates the $(k - 2, k)$ entry and creates a new nonzero entry in position $(k - 3, k)$. Subsequent rotators push the extra nonzero entry up the k th column until it disappears off the top of the matrix. The resulting bidiagonal matrix has both $\beta_k = 0$ and $\gamma_{k-1} = 0$.
- Supposing still that $\beta_k = 0$, show that we can make $\gamma_k = 0$ by a sequence of $m - k$ rotators applied to B from the left. The procedure is analogous to the procedure of part (b), except that it pushes the unwanted nonzero entry out along the k th row rather than up the k th column.

Once the procedures of parts (b) and (c) have been applied, we have $\beta_k = 0$, $\gamma_{k-1} = 0$, and $\gamma_k = 0$. The bidiagonal matrix now has the form

$$\begin{bmatrix} B_1 & & \\ & 0 & \\ & & B_2 \end{bmatrix},$$

where the zero that is shown explicitly is β_k . This is a zero singular value of B . We can move it to the (m, m) position of the matrix by a row and column interchange, if we wish. The rest of the SVD can be found by computing the SVDs of B_1 and B_2 separately and then combining them. $\square$

Exercise 5.9.47 Show that if B is properly bidiagonal, then there exist orthogonal, diagonal matrices D_1 and D_2 (these have entries ± 1 on the main diagonal), such that all of the nonzero entries of $D_1 B D_2$ are positive. Thus we can assume, without loss of generality, that all β_k and γ_k are positive. $\square$

Exercise 5.9.48 Show that if $A \in \mathbb{R}^{n \times n}$ is a symmetric, positive definite matrix, then its singular values are the same as its eigenvalues, and its right and left singular vectors and (normalized) eigenvectors are all the same. How is the picture changed if A is symmetric but indefinite? $\square$

Exercise 5.9.49 Given $A \in \mathbb{R}^{n \times m}$, consider the symmetric matrix

$$M = \begin{bmatrix} 0 & A^T \\ A & 0 \end{bmatrix} \in \mathbb{R}^{(n+m) \times (n+m)}.$$

Show that a simple relationship exists between the singular vectors of A and the eigenvectors of M . Show how to build an orthogonal basis of $\mathbb{R}^{n+m}$ consisting of eigenvectors of M , given the singular vectors of A . In the process, find all of the eigenvalues of M . (*Hint:* If μ is an eigenvalue of M , then so is $-\mu$.) $\square$

Exercise 5.9.50 Let $A \in \mathbb{R}^{n \times m}$ and $C \in \mathbb{R}^{m \times n}$, and define

$$F = \begin{bmatrix} 0 & C \\ A & 0 \end{bmatrix} \in \mathbb{R}^{(n+m) \times (n+m)}.$$

Show that μ is an eigenvalue of F if and only if μ^2 is an eigenvalue of both AC and CA . Determine the relationship between eigenvectors of M and eigenvectors of AC and CA . $\square$

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6

Eigenvalues and Eigenvectors II

The first section of this chapter introduces the important notions of *eigenspace* and *invariant subspace*. The latter concept is particularly important for a deeper understanding of eigenvalue problems and algorithms. Armed with this new concept, we are able to explain in Section 6.2 why the QR algorithm works. In the process we introduce *simultaneous iteration*, an algorithm that is closely related to QR and is also useful in its own right for computing eigenvalues and eigenvectors of large, sparse matrices. Although the importance of simultaneous iteration as a practical algorithm has declined in recent years, it does nevertheless provide us with a smooth entry into Sections 6.3 and 6.4, which discuss several algorithms for large, sparse eigenvalue problems.

The last three sections of the chapter can be read more or less independently. Section 6.5 discusses the sensitivity of eigenvalues and eigenvectors, and introduces condition numbers for these objects. Section 6.6 surveys a variety of methods that can be used as alternatives to the QR algorithm on symmetric matrices. Some of these methods are superior to QR for symmetric, tridiagonal eigenvalue problems. Finally, Section 6.7 provides a brief introduction to the *generalized eigenvalue problem* $Av = \lambda Bv$ and algorithms for solving it.

6.1 EIGENSPACES AND INVARIANT SUBSPACES

Let $A \in \mathbb{C}^{n \times n}$, let λ be any complex number, and consider the set of vectors

$$\mathcal{S}_\lambda = \{v \in \mathbb{C}^n \mid Av = \lambda v\}.$$

You can easily check that $\mathcal{S}_\lambda$ is a subspace of $\mathbb{C}^n$.

Exercise 6.1.1 (a) Show that if $v_1, v_2 \in \mathcal{S}_\lambda$, then $v_1 + v_2 \in \mathcal{S}_\lambda$. (b) Show that if $v \in \mathcal{S}_\lambda$ and $\alpha \in \mathbb{C}$, then $\alpha v \in \mathcal{S}_\lambda$. Thus $\mathcal{S}_\lambda$ is a subspace of $\mathbb{C}^n$. $\square$

Since the equation $Av = \lambda v$ can be rewritten as $(\lambda I - A)v = 0$, we see that $\mathcal{S}_\lambda = \mathcal{N}(\lambda I - A)$, the null space of $\lambda I - A$. The space $\mathcal{S}_\lambda$ always contains the zero vector. It contains other vectors if and only if λ is an eigenvalue of A , in which case we call $\mathcal{S}_\lambda$ the *eigenspace* of A associated with λ .

The notion of an invariant subspace is a useful generalization of eigenspace. Since we will want to talk about both real and complex invariant subspaces, we will begin by introducing some notation to make the presentation more flexible. Let $\mathbb{F}$ denote either the real or complex number field. Thus $\mathbb{F}^n$ and $\mathbb{F}^{n \times n}$ will denote either $\mathbb{R}^n$ and $\mathbb{R}^{n \times n}$ or $\mathbb{C}^n$ and $\mathbb{C}^{n \times n}$, depending on the choice of $\mathbb{F}$.

Let $A \in \mathbb{F}^{n \times n}$. A subspace $\mathcal{S}$ of $\mathbb{F}^n$ is said to be *invariant* under A if $Ax \in \mathcal{S}$ whenever $x \in \mathcal{S}$, in other words, $A\mathcal{S} \subseteq \mathcal{S}$. We will also sometimes bend the language slightly and say that $\mathcal{S}$ is an *invariant subspace* of A . The spaces $\{0\}$ and $\mathbb{F}^n$ are both invariant subspaces of A . The next exercise gives some nontrivial examples of invariant subspaces.

First we recall some notation from Chapter 3. Given $x_1, x_2, \dots, x_k \in \mathbb{F}^m$, $\text{span}\{x_1, \dots, x_k\}$ denotes the set of all linear combinations of $x_1, \dots, x_k$. That is,

$$\text{span}\{x_1, \dots, x_k\} = \left\{ \sum_{i=1}^k c_i x_i \mid c_i \in \mathbb{F}, i = 1, \dots, k \right\}.$$

Notice that there is now some ambiguity in the meaning of $\text{span}\{x_1, \dots, x_k\}$. The coefficients c_i are taken from $\mathbb{F}$. They can be either real or complex, depending on the context. There was no such ambiguity in Chapter 3, because there we always had $\mathbb{F} = \mathbb{R}$.

Exercise 6.1.2

- (a) Let $v \in \mathbb{F}^n$ be any eigenvector of A . Show that $\text{span}\{v\}$ is invariant under A .
- (b) Let $\mathcal{S}_\lambda$ be any eigenspace of A (of dimension 1 or greater). Show that $\mathcal{S}_\lambda$ is invariant under A .
- (c) Let $v_1, \dots, v_k \in \mathbb{F}^n$ be any k eigenvectors of A associated with eigenvalues $\lambda_1, \dots, \lambda_k$. Show that the space $\mathcal{S} = \text{span}\{v_1, \dots, v_k\}$ is invariant under A . It can be shown that if A is semisimple, every invariant subspace has this form.
- (d) Let

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{bmatrix}.$$

Find the eigenspaces of A , and find a space $\mathcal{S}$ that is invariant under A and is not spanned by eigenvectors of A .

$\square$

If we think of A as a linear operator mapping $\mathbb{F}^n$ into $\mathbb{F}^n$, then an invariant subspace can be used to reduce the operator, in the following sense. Suppose $\mathcal{S}$ is an invariant subspace under A that is neither $\{0\}$ nor $\mathbb{F}^n$, and let $\hat{A}$ denote the restricted operator that acts on the subspace $\mathcal{S}$, that is, $\hat{A} = A|_{\mathcal{S}}$. Then $\hat{A}$ maps $\mathcal{S}$ into $\mathcal{S}$. It is clear the every eigenvector of $\hat{A}$ is an eigenvector of A , and so is every eigenvalue. Thus we can obtain information about the eigensystem of A by studying $\hat{A}$, an operator that is simpler than A in the sense that it acts on a lower-dimensional space. The following sequence of theorems and exercises shows how this reduction can be carried out in practice. We begin by characterizing invariant subspaces in the language of matrices.

Theorem 6.1.3 *Let $\mathcal{S}$ be a subspace of $\mathbb{F}^n$ with a basis $x_1, \dots, x_k$. Thus $\mathcal{S} = \text{span}\{x_1, \dots, x_k\}$. Let $\hat{X} = \begin{bmatrix} x_1 & \cdots & x_k \end{bmatrix} \in \mathbb{F}^{n \times k}$. Then $\mathcal{S}$ is invariant under $A \in \mathbb{F}^{n \times n}$ if and only if there exists $\hat{B} \in \mathbb{F}^{k \times k}$ such that*

$$A\hat{X} = \hat{X}\hat{B}.$$

Proof. Suppose there is a $\hat{B}$ such that $A\hat{X} = \hat{X}\hat{B}$. Let b_{ij} denote the (i, j) entry of $\hat{B}$. Equating the j th column of $A\hat{X}$ with the j th column of $\hat{X}\hat{B}$, we see that $Ax_j = \sum_{i=1}^k x_i b_{ij} \in \text{span}\{x_1, \dots, x_k\} \in \mathcal{S}$. Since $Ax_j \in \mathcal{S}$ for $j = 1, \dots, k$, and $x_1, \dots, x_k$ span $\mathcal{S}$, it follows that $Ax \in \mathcal{S}$ for all $x \in \mathcal{S}$. Thus $\mathcal{S}$ is invariant under A . conversely, assume $\mathcal{S}$ is invariant. We must construct $\hat{B}$ such that $A\hat{X} = \hat{X}\hat{B}$. For $j = 1, \dots, k$, $Ax_j \in \mathcal{S}$, so there exist constants $b_{1j}, \dots, b_{kj} \in \mathbb{F}$ such that $Ax_j = x_1 b_{1j} + x_2 b_{2j} + \cdots + x_k b_{kj}$. Define $\hat{B} \in \mathbb{F}^{k \times k}$ to be the matrix whose (i, j) entry is b_{ij} . Then $A[x_1 \cdots x_k] = [x_1 \cdots x_k]\hat{B}$, that is, $A\hat{X} = \hat{X}\hat{B}$. (See also Exercise 6.1.12.) $\square$

Exercise 6.1.4 Let $\mathcal{S}$ be an invariant subspace of A with basis $x_1, \dots, x_k$, and let $\hat{X} = [x_1 \cdots x_k]$. By Theorem 6.1.3 there exists $\hat{B} \in \mathbb{F}^{k \times k}$ such that $A\hat{X} = \hat{X}\hat{B}$.

- (a) Show that if $\hat{v} \in \mathbb{F}^k$ is an eigenvector of $\hat{B}$ with eigenvalue λ , then $v = \hat{X}\hat{v}$ is an eigenvector of A with eigenvalue λ . In particular, every eigenvalue of $\hat{B}$ is an eigenvalue of A . What role does the linear independence of $x_1, \dots, x_k$ play?

- (b) Show that $v \in \mathcal{S}$. Thus v is an eigenvector of $A|_{\mathcal{S}}$.

$\square$

Exercise 6.1.5 With A , $\hat{B}$, and $\mathcal{S}$ as in the previous exercise, show that $\hat{B}$ is a matrix representation of the linear operator $A|_{\mathcal{S}}$. $\square$

The next theorem shows that whenever we have (a basis for) a nontrivial invariant subspace, we can make a similarity transformation that breaks the eigenvalue problem into two smaller problems.

Theorem 6.1.6 *Let $\mathcal{S}$ be invariant under $A \in \mathbb{F}^{n \times n}$. Suppose $\mathcal{S}$ has dimension k , with $1 \leq k < n$, and let $x_1, \dots, x_k$ be a basis of $\mathcal{S}$. Let $x_{k+1}, \dots, x_n$ be any*

$n - k$ vectors such that the vectors $x_1, \dots, x_n$ together form a basis for $\mathbb{F}^n$. Let $X_1 = [x_1 \cdots x_k]$, $X_2 = [x_{k+1} \cdots x_n]$, and $X = [x_1 \cdots x_n] \in \mathbb{F}^{n \times n}$. Define $B = X^{-1}AX$. Then B is block upper triangular:

$$B = \begin{bmatrix} B_{11} & B_{12} \\ 0 & B_{22} \end{bmatrix},$$

where $B_{11} \in \mathbb{F}^{k \times k}$. Furthermore $AX_1 = X_1B_{11}$.

B is similar to A , so it has the same spectrum. This can be found by computing the spectra of B_{11} and B_{22} separately. Thus the eigenvalue problem has been broken into two smaller problems.

Notice that B_{11} is the same as the matrix $\hat{B}$ that was introduced in Theorem 6.1.3.

If we wish, we can always take $x_1, \dots, x_k$ to be an orthonormal basis of $\mathcal{S}$ and choose the additional vectors $x_{k+1}, \dots, x_n$ so that $x_1, \dots, x_n$ together form an orthonormal basis of $\mathbb{F}^n$. This can be achieved by the Gram-Schmidt process, for example. Then X is unitary, and B is unitarily similar to A .

Proof. The equation $B = X^{-1}AX$ is equivalent to $AX = XB$. The j th column of this equation is $Ax_j = \sum_{i=1}^n x_i b_{ij}$. Since $x_1, \dots, x_n$ are linearly independent, this sum is the unique representation of Ax_j as a linear combination of $x_1, \dots, x_n$. On the other hand, for $j = 1, \dots, k$, x_j lies in the invariant subspace $\mathcal{S} = \text{span}\{x_1, \dots, x_k\}$, so Ax_j lies in $\mathcal{S}$ as well. Thus $Ax_j = c_{1j}x_1 + \cdots + c_{kj}x_k$ for some $c_{1j}, \dots, c_{kj} \in \mathbb{F}^n$. By uniqueness of the representation of Ax_j , we have $b_{ij} = c_{ij}$ for $i = 1, \dots, k$, and more importantly $b_{ij} = 0$ for $i = k+1, \dots, n$. This holds for $j = 1, \dots, k$. In the partition $B = \begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix}$, the block B_{21} consists precisely of those b_{ij} for which $k+1 \leq i \leq n$ and $1 \leq j \leq k$. Thus $B_{21} = 0$. This proves that B is block triangular. The equation $AX_1 = X_1B_{11}$ now follows immediately from the obvious partition of the equation $AX = XB$. $\square$

See Exercise 6.1.13 for a second proof of this theorem.

Exercise 6.1.7 Prove the converse of Theorem 6.1.6: if $B_{21} = 0$, then $\mathcal{S}$ is invariant under A . $\square$

Corollary 6.1.8 Let $B = X^{-1}AX$, where $X = [x_1 \cdots x_n] \in \mathbb{F}^{n \times n}$ is nonsingular. Then B is upper triangular if and only if all of the spaces

$$\text{span}\{x_1, \dots, x_k\}, \quad k = 1, \dots, n-1$$

are invariant under A .

Exercise 6.1.9 Prove Corollary 6.1.8. $\square$

Theorem 6.1.6 shows that whenever we find a nontrivial invariant subspace, we can make a similarity transformation that breaks the eigenvalue problem into two smaller problems. If we can then find subspaces that are invariant under B_{11} and

B_{22} , we can break each of them into smaller subproblems. If we can continue this process indefinitely, we can eventually break the problem down to many small submatrices of size 1×1 and 2×2 , from which we can read off the eigenvalues. It is thus reasonable to say that finding invariant subspaces is the fundamental task of eigensystem computations.

Schur's Theorem (Theorem 5.4.11) and its real variant, the Wintner-Murnaghan Theorem (Theorem 5.4.22) show that the breakdown process described in the previous paragraph is always possible in principle. We now see (cf. Corollary 6.1.8) that these are theorems about invariant subspaces.

The QR algorithm achieves the practical task of breaking the matrix down. In a typical reduction the submatrix B_{22} is tiny and B_{11} is huge. Thus (nearly) all of the remaining work consists of breaking down B_{11} . It would be nice to develop an algorithm that breaks a matrix into submatrices of roughly equal size. This requires the identification of an invariant subspace of dimension approximately $n/2$, which has proved to be a difficult task.¹

Additional Exercises

Exercise 6.1.10 If μ is an eigenvalue of A , then the eigenspace $S_\mu(A) = \{v \in \mathbb{C}^n \mid Av = \mu v\}$ has dimension at least 1. Often the dimension is exactly 1, but sometimes it is greater. The *geometric multiplicity* of the eigenvalue μ is defined to be the dimension of the eigenspace $S_\mu(A)$. The geometric multiplicity may be different from the *algebraic multiplicity*, which is defined to be the number of times μ appears as a root of the characteristic equation. By this we mean the number of times the factor $\lambda - \mu$ appears as a factor of the characteristic polynomial $\det(\lambda I - A)$.

- (a) Find the algebraic and geometric multiplicities of the eigenvalue $\lambda = 2$ in each of the following matrices.

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & 3 \\ 0 & 0 & 3 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{bmatrix} \quad C = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

- (b) Schur's Theorem (Theorem 5.4.11) guarantees that every matrix $A \in \mathbb{C}^{n \times n}$ is unitarily similar to an upper triangular matrix T : $A = UTU^{-1}$. Show that the algebraic multiplicity of μ as an eigenvalue of A is equal to the number of times μ appears on the main diagonal of T .
- (c) Show that if A and B are similar matrices, then $\dim S_\mu(A) = \dim S_\mu(B)$. Thus the geometric multiplicity of μ as an eigenvalue of A is the same as it is as an eigenvalue of B .
- (d) Show that the geometric multiplicity of μ is less than or equal to the algebraic multiplicity. (Work with the upper triangular matrix T , and suppose the alge-

¹ However, some headway has been made in this area. See [4].

braic multiplicity of μ is k . We know that in Schur's Theorem we can arrange the eigenvalues in any order on the main diagonal of T (Exercise 5.4.30). Suppose T is constructed so that its first k main diagonal entries are μ . What does an eigenvector look like? Show that the dimension of the eigenspace cannot exceed k .)

□

Exercise 6.1.11 (Proof of Wintner-Murnaghan Theorem 5.4.22)

- Prove that if $A \in \mathbb{R}^{n \times n}$ with $n \geq 3$, then there is a subspace $S \in \mathbb{R}^n$ of dimension 1 or 2 that is invariant under A . (Consider an eigenvector of A . Its real and imaginary parts span a subspace of $\mathbb{R}^n$. Show that this space is invariant and has dimension 1 or 2.)
- Using part (a) and Theorem 6.1.6, prove Theorem 5.4.22 by induction on n .

□

Exercise 6.1.12 Let $x_1, \dots, x_k \in \mathbb{F}^n$, $\hat{X} = [x_1 \cdots x_k] \in \mathbb{F}^{n \times k}$, and $S = \text{span}\{x_1, \dots, x_k\} \subseteq \mathbb{F}^n$. Viewing $\hat{X}$ as a linear operator mapping $\mathbb{F}^k$ into $\mathbb{F}^n$, the *range* of $\hat{X}$ (cf. Section 3.5) is defined by $\mathcal{R}(\hat{X}) = \{\hat{X}z \mid z \in \mathbb{F}^k\}$.

- Show that $\mathcal{R}(\hat{X}) = S$.
- Using the observation from part (a), write a shorter proof of Theorem 6.1.3.

□

Exercise 6.1.13 This exercise derives a second proof of Theorem 6.1.6. Assuming the same notation as in Theorem 6.1.6, define $Y \in \mathbb{F}^{n \times n}$ by $Y^T = X^{-1}$. Let $y_1, \dots, y_n$ denote the columns of Y , and let $Y_1 = [y_1 \cdots y_k]$ and $Y_2 = [y_{k+1} \cdots y_n]$.

- Show that $Y_1^T X_1 = I \in \mathbb{F}^{k \times k}$, $Y_2^T X_2 = I \in \mathbb{F}^{(n-k) \times (n-k)}$, $Y_2^T X_1 = 0$, and $Y_1^T X_2 = 0$.
- Show that $Y_1^T A X_1 = B_{11}$, $Y_2^T A X_2 = B_{22}$, $Y_2^T A X_1 = B_{21}$, and $Y_1^T A X_2 = B_{12}$.
- Use Theorem 6.1.3 and the selected equations from parts (a) and (b) to show that $B_{21} = 0$, if the columns of X_1 span an invariant subspace.

□

Exercise 6.1.14 In the notation of Theorem 6.1.6, let $\mathcal{T} = \text{span}\{x_{k+1}, \dots, x_n\}$.

- Show that $\mathcal{T}$ is invariant under A if and only if $B_{12} = 0$. Your solution may resemble the proof of Theorem 6.1.6, or you can use ideas from Exercise 6.1.13.
- Suppose $\mathcal{T}$ is invariant under A . Show that if $\hat{v}$ is an eigenvector of B_{22} associated with the eigenvalue λ , then $v = X_2 \hat{v}$ is an eigenvector of A associated with eigenvalue λ .

□

Exercise 6.1.15 Suppose the hypotheses of Theorem 6.1.6 are satisfied. Define $Y \in \mathbb{F}^{n \times n}$ by $Y^T = X^{-1}$, let $y_1, \dots, y_n$ denote the columns of Y , let $Y_1 = [y_1 \cdots y_k]$, $Y_2 = [y_{k+1} \cdots y_n]$, and define $\mathcal{U} = \text{span}\{y_{k+1}, \dots, y_n\}$.

- Show that $\mathcal{U}$ is invariant under A^T .
- Show that if $\hat{w}$ is an eigenvector of B_{22}^T associated with the eigenvalue λ , then $w = Y_2 \hat{w}$ is an eigenvector of A^T associated with λ .

□

Exercise 6.1.16 We repeat Exercise 6.1.15 with one minor alteration. Take $\mathbb{F} = \mathbb{C}$ and define $Z \in \mathbb{C}^{n \times n}$ by $Z^* = X^{-1}$. Let $z_1, \dots, z_n$ denote the columns of Z , and define $Z_2 = [z_{k+1} \cdots z_n]$ and $\mathcal{V} = \text{span}\{z_{k+1}, \dots, z_n\}$.

- Show that $\mathcal{V}$ is invariant under A^* .
- Show that if $\hat{w}$ is an eigenvector of B_{22}^* associated with the eigenvalue $\bar{\lambda}$, then $w = Z_2 \hat{w}$ is an eigenvector of A^* associated with $\bar{\lambda}$.
- It is always possible to choose $x_1, \dots, x_n$ so that X is unitary, in which case $Z = X$. Use these facts to show that if $\mathcal{S}$ is invariant under A , then $\mathcal{S}^\perp$ is invariant under A^* .

□

Exercise 6.1.17 In this exercise we prove directly (without using bases) that $\mathcal{S}^\perp$ is invariant under A^* if $\mathcal{S}$ is invariant under A . Recall that the inner product in $\mathbb{C}^n$ is defined by $\langle x, y \rangle = \sum_{j=1}^n x_j \bar{y}_j$. Let $A \in \mathbb{C}^{n \times n}$.

- Show that for all $x, y \in \mathbb{C}^n$, $\langle Ax, y \rangle = \langle x, A^*y \rangle$. This is the complex analogue of Lemma 3.5.9.
- Use the result of part (a) to prove directly that if $\mathcal{S}$ is invariant under A , then $\mathcal{S}^\perp$ is invariant under A^* .
- Conclude from part (b) that if A is Hermitian ($A = A^*$), then $\mathcal{S}$ is invariant under A if and only if $\mathcal{S}^\perp$ is. Relate this result to Exercise 6.1.14 and Theorem 6.1.6.

□

Exercise 6.1.18 Let $A, B, X \in \mathbb{F}^{n \times n}$ with X nonsingular and $B = X^{-1}AX$. Let $x_1, \dots, x_n$ denote the columns of X . Prove that B is lower triangular if and only if the spaces $\text{span}\{x_{k+1}, \dots, x_n\}$, $k = 1, \dots, n-1$, are all invariant under A .

□

6.2 SUBSPACE ITERATION, SIMULTANEOUS ITERATION, AND THE QR ALGORITHM

The principal aims of this section are to derive the QR algorithm in a natural, logical manner and to explain its convergence properties. Along the way we will encounter subspace iteration and simultaneous iteration, concepts that are also important in connection with calculation of eigenvalues of large, sparse matrices. We will also discuss an important duality principle, which states that whenever direct (power) iteration takes place, inverse iteration automatically takes place at the same time. This principle will allow us to connect the QR algorithm with inverse iteration and Rayleigh quotient iteration.

Subspace Iteration

We begin by returning to the power method and recasting it in terms of subspaces. Given a matrix $A \in \mathbb{C}^{n \times n}$ that has a dominant eigenvalue, we can use the power method to calculate a dominant eigenvector v_1 or a multiple thereof. It does not matter which multiple we obtain, since each nonzero multiple is as good an eigenvector as any other. In fact each multiple of v_1 is just a representative of the eigenspace $\text{span}\{v_1\}$, which is the real object of interest. Likewise, in the power sequence $q, Aq, A^2q, A^3q, \dots$ each of the iterates $A^m q$ can be viewed as a representative of the space $\text{span}\{A^m q\}$, which it spans. The operation of rescaling a vector amounts to replacing one representative by another representative of the same one-dimensional space. Thus the power method can be viewed as a process of iterating on subspaces: First a one-dimensional subspace $S = \text{span}\{q\}$ is chosen. Then the iterates

$$S, AS, A^2S, A^3S, \dots \quad (6.2.1)$$

are formed. This sequence converges linearly to the eigenspace $\mathcal{T} = \text{span}\{v_1\}$ in the sense that the angle between $A^m S$ and $\mathcal{T}$ converges to zero as $m \rightarrow \infty$.

It is quite natural to generalize this process to subspaces of dimension greater than one. Thus we can choose a subspace S of any dimension and form the sequence (6.2.1). It is perhaps not surprising that this sequence will generally converge to a space that is invariant under A . Before proceeding, you should work the following exercise, which reviews a number of concepts and covers the most basic properties of subspace iteration.

Exercise 6.2.2 Let $A \in \mathbb{F}^{n \times n}$, where $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$, and let S be a subspace of $\mathbb{F}^n$. then AS is defined by $AS = \{Ax \mid x \in S\}$.

- Recall that a nonempty subset $\mathcal{U}$ of $\mathbb{F}^n$ is a *subspace* if and only if (i) $x_1, x_2 \in \mathcal{U} \Rightarrow x_1 + x_2 \in \mathcal{U}$ and (ii) $\alpha \in \mathbb{F}$ and $x \in \mathcal{U} \Rightarrow \alpha x \in \mathcal{U}$. Show that if S is a subspace of $\mathbb{F}^n$, then AS is also a subspace of $\mathbb{F}^n$.
- By definition, $A^m S = \{A^m x \mid x \in S\}$. Show that $A^m S = A(A^{m-1}S)$.
- Show that if $S = \text{span}\{x_1, \dots, x_k\}$, then $AS = \text{span}\{Ax_1, \dots, Ax_k\}$.

- (d) Recall that the *null space* of A is $\mathcal{N}(A) = \{x \in \mathbb{F}^n \mid Ax = 0\}$. Let S be a subspace of $\mathbb{F}^n$ for which $S \cap \mathcal{N}(A) = \{0\}$. Show that if $x_1, \dots, x_k$ is a basis of S , then $Ax_1, \dots, Ax_k$ is a basis of AS . Consequently $\dim(AS) = \dim(S)$.

□

In order to talk about convergence of subspaces, we need to say what we mean by the distance between two subspaces. However, a detailed discussion would distract us from our current task, which is to give you an intuitive feel for why the QR algorithm works. We will therefore just say that the *distance* between two subspaces S_1 and S_2 , denoted $d(S_1, S_2)$, is defined to be the sine of the largest principal angle between them. (Principal angles are defined in [33], for example.) Given a sequence of subspaces (S_m) and a subspace $\mathcal{T}$, all of the same dimension, we will say that (S_m) *converges* to $\mathcal{T}$ (denoted symbolically by $S_m \rightarrow \mathcal{T}$) provided that $d(S_m, \mathcal{T}) \rightarrow 0$ as $m \rightarrow \infty$. We are now ready to state the main theorem on the convergence of subspace iteration.

Theorem 6.2.3 *Let $A \in \mathbb{F}^{n \times n}$ be semisimple with linearly independent eigenvectors $v_1, \dots, v_n \in \mathbb{F}^n$ and associated eigenvalues $\lambda_1, \dots, \lambda_n \in \mathbb{F}$, satisfying $|\lambda_1| \geq |\lambda_2| \geq \dots \geq |\lambda_n|$. Suppose $|\lambda_k| > |\lambda_{k+1}|$ for some k . Let $\mathcal{T}_k = \text{span}\{v_1, \dots, v_k\}$ and $\mathcal{U}_k = \text{span}\{v_{k+1}, \dots, v_n\}$. Let S be any k -dimensional subspace of $\mathbb{F}^n$ such that $S \cap \mathcal{U}_k = \{0\}$. Then there exists a constant C such that*

$$d(A^m S, \mathcal{T}_k) \leq C |\lambda_{k+1}/\lambda_k|^m, \quad m = 0, 1, 2, \dots$$

Thus $A^m S \rightarrow \mathcal{T}_k$ linearly with convergence ratio $|\lambda_{k+1}/\lambda_k|$.

Since $\mathcal{T}_k$ is spanned by eigenvectors of A , it is invariant under A . $\mathcal{T}_k$ is called the *dominant* invariant subspace of A of dimension k . The semisimplicity assumption is not crucial; a slightly revised version of Theorem 6.2.3 holds for defective matrices. For a proof see [79].

Exercise 6.2.4

- (a) Show that the condition $|\lambda_k| > |\lambda_{k+1}|$ implies that $\mathcal{N}(A) \subseteq \mathcal{U}_k$.
 (b) More generally, show that $\mathcal{N}(A^m) \subseteq \mathcal{U}_k$ for all $m > 0$.
 (c) Conclude that $A^m S$ has dimension k for all $m > 0$.

□

It is easy to argue the plausibility of Theorem 6.2.3. Let q be any nonzero vector in S . We can easily show that the iterates $A^m q$ lie (relatively) close and closer to $\mathcal{T}_k$ as m increases. Indeed q may be expressed uniquely in the form

$$\begin{aligned} q &= c_1 v_1 + c_2 v_2 + \dots + c_k v_k && \text{(component in } \mathcal{T}_k) \\ &+ c_{k+1} v_{k+1} + \dots + c_n v_n. && \text{(component in } \mathcal{U}_k) \end{aligned}$$

Since $q \notin \mathcal{U}_k$, at least one of the coefficients $c_1, \dots, c_k$ must be nonzero. Now

$$\begin{aligned} A^m q / (\lambda_k)^m &= c_1 (\lambda_1 / \lambda_k)^m v_1 + \cdots + c_{k-1} (\lambda_{k-1} / \lambda_k)^m v_{k-1} + c_k v_k \\ &\quad + c_{k+1} (\lambda_{k+1} / \lambda_k)^m v_{k+1} + \cdots + c_n (\lambda_n / \lambda_k)^m v_n. \end{aligned}$$

The coefficients of the component in $\mathcal{T}_k$ increase, or at least do not decrease, as m increases. In contrast, the coefficients of the component in $\mathcal{U}_k$ tend to zero with rate $|\lambda_{k+1}/\lambda_k|$ or better. Thus each sequence $(A^m q)$ converges to $\mathcal{T}_k$ at the stated rate or better, and consequently the limit of $(A^m \mathcal{S})$ lies in $\mathcal{T}_k$. The limit cannot be a proper subspace of $\mathcal{T}_k$ because it has dimension k .

The hypothesis $\mathcal{S} \cap \mathcal{U}_k = \{0\}$ merits some comment. Is this a stringent condition or is it not? First let us see what it amounts to in the case $k = 1$. In this case we have $\mathcal{S} = \text{span}\{q\}$, and $\mathcal{U}_1 = \text{span}\{v_2, \dots, v_n\}$. Clearly $\mathcal{S} \cap \mathcal{U}_1 = \{0\}$ if and only if $q \notin \mathcal{U}_1$. This just means that $c_1 \neq 0$ in the unique expansion $q = c_1 v_1 + c_2 v_2 + \cdots + c_n v_n$. You will recall from Section 5.3 that this is the condition for convergence of the basic power method to $\text{span}\{v_1\}$. It is satisfied by almost every $v \in \mathbb{F}^n$. In the language of subspaces, a one-dimensional subspace $\mathcal{S}$ and an $(n-1)$ -dimensional subspace $\mathcal{U}_1$ are almost certain to satisfy $\mathcal{S} \cap \mathcal{U}_1 = \{0\}$. Fortunately the same is true in general, as long as the sum of the dimensions of $\mathcal{S}$ and $\mathcal{U}_k$ does not exceed n .

To get a feel for why this might be so, note first that if the sum of the dimensions *does* exceed n , then $\mathcal{S}$ and $\mathcal{U}_k$ must intersect nontrivially. This is a consequence of the fundamental relationship

$$\dim(\mathcal{S} \cap \mathcal{U}_k) + \dim(\mathcal{S} + \mathcal{U}_k) = \dim(\mathcal{S}) + \dim(\mathcal{U}_k), \quad (6.2.5)$$

since $\dim(\mathcal{S} + \mathcal{U}_k) \leq \dim(\mathbb{F}^n) = n$.

In our present situation $\dim(\mathcal{S}) = k$ and $\dim(\mathcal{U}_k) = n - k$. The dimensions add to n exactly, so there is enough room in $\mathbb{F}^n$ that $\mathcal{S}$ and $\mathcal{U}_k$ are not forced by (6.2.5) to intersect nontrivially. If two subspaces are not forced to intersect nontrivially, then they almost certainly will not. Consider the situation in $\mathbb{R}^3$. There any two two-dimensional subspaces (planes through the origin) are required to intersect nontrivially because the sum of their dimensions exceeds three. In contrast, a plane and a line are not required to intersect nontrivially, and it is obvious that they almost certainly will not. See also Exercise 6.2.33. We conclude that the condition $\mathcal{S} \cap \mathcal{U}_k = \{0\}$ of Theorem 6.2.3 is not stringent; it is satisfied by practically any $\mathcal{S}$ chosen at random.

Simultaneous Iteration

In order to carry out subspace iteration in practice, we must choose a basis for $\mathcal{S}$ and iterate on the basis vectors simultaneously. Let $q_1^{(0)}, q_2^{(0)}, \dots, q_k^{(0)}$ be a basis for $\mathcal{S}$. From Exercises 6.2.2 and 6.2.4 we know that if $\mathcal{S} \cap \mathcal{U}_k = \{0\}$, then $A^m q_1^{(0)}, A^m q_2^{(0)}, \dots, A^m q_k^{(0)}$ form a basis for $A^m \mathcal{S}$. Thus, in theory, we can simply iterate

on a basis of $\mathcal{S}$ to obtain bases for $A\mathcal{S}$, $A^2\mathcal{S}$, $A^3\mathcal{S}$ and so on. There are two reasons why it is not advisable to do this in practice: (i) The vectors will have to be rescaled regularly in order to avoid underflow or overflow. (ii) Each of the sequences $q_i^{(0)}$, $Aq_i^{(0)}$, $A^2q_i^{(0)}$, $\dots$ independently converges to the dominant eigenspace $\text{span}\{v_1\}$ (assuming $|\lambda_1| > |\lambda_2|$). It follows that for large m the vectors $A^mq_1^{(0)}, \dots, A^mq_k^{(0)}$ all point in nearly the same direction. Thus they form an ill-conditioned basis of $A^m\mathcal{S}$. In numerical practice, an ill-conditioned basis determines the space poorly; a small perturbation in one of the basis vectors can make a big change in the space.

Ill-conditioned bases can be avoided by replacing the basis obtained at each step by a well-conditioned basis for the same subspace. This replacement operation can also incorporate the necessary rescaling. There are numerous ways to do this, but the most reliable is to orthonormalize. Thus the following *simultaneous iteration* procedure is recommended.

$$\begin{array}{l} \text{Start with an orthonormal basis } q_1^{(0)}, \dots, q_k^{(0)} \text{ of } \mathcal{S} \\ \text{for } m = 0, 1, 2, \dots \\ \left\{ \begin{array}{l} \text{Calculate } Aq_1^{(m)}, \dots, Aq_k^{(m)}, \text{ a basis for } A^{m+1}\mathcal{S} \\ \text{Orthonormalize } Aq_1^{(m)}, \dots, Aq_k^{(m)} \\ \text{to obtain } q_1^{(m+1)}, \dots, q_k^{(m+1)} \end{array} \right. \end{array} \quad (6.2.6)$$

In practice one should use a robust process to perform the orthonormalization. For example, one could use modified Gram-Schmidt with reorthogonalization or a QR decomposition by reflectors (See Section 3.4).

The simultaneous iteration procedure (6.2.6) has the agreeable property of iterating on lower-dimensional subspaces at no extra cost. For $i = 1, \dots, k$, let $\mathcal{S}_i$ denote the i -dimensional subspace spanned by $q_1^{(0)}, \dots, q_i^{(0)}$. Then

$$A\mathcal{S}_i = \text{span}\{Aq_1^{(0)}, \dots, Aq_i^{(0)}\} = \text{span}\{q_1^{(1)}, \dots, q_i^{(1)}\},$$

by the subspace-preserving property (3.4.12) of the Gram-Schmidt procedure. In general

$$A^m\mathcal{S}_i = \text{span}\{q_1^{(m)}, \dots, q_i^{(m)}\},$$

so $\text{span}\{q_1^{(m)}, \dots, q_i^{(m)}\}$ converges to the invariant subspace $\text{span}\{v_1, \dots, v_i\}$ as $m \rightarrow \infty$, provided that appropriate hypotheses are satisfied. Thus simultaneous iteration seeks not only an invariant subspace of dimension k , but subspaces of dimensions $1, 2, \dots, k-1$ as well.

Although simultaneous iteration is an interesting algorithm in its own right, our main reason for introducing it here is to use it as a vehicle for the introduction of the QR algorithm. To that end, let us now consider what happens when simultaneous iteration is applied to a complete set of orthonormal vectors $q_1^{(0)}, \dots, q_n^{(0)} \in \mathbb{F}^{n \times n}$. We continue to assume that A is semisimple, with linearly independent eigenvectors $v_1, \dots, v_n$. For $k = 1, \dots, n-1$, let $\mathcal{S}_k = \text{span}\{q_1^{(0)}, \dots, q_k^{(0)}\}$, $\mathcal{T}_k =$

$\text{span}\{v_1, \dots, v_k\}$, and $\mathcal{U}_k = \text{span}\{v_{k+1}, \dots, v_n\}$, and assume $S_k \cap \mathcal{U}_k = \{0\}$ and $|\lambda_k| > |\lambda_{k+1}|$. Then (for $k = 1, \dots, n-1$) $A^m S_k = \text{span}\{q_1^{(m)}, \dots, q_k^{(m)}\} \rightarrow \mathcal{T}_k$ linearly with convergence ratio $|\lambda_{k+1}/\lambda_k|$ as $m \rightarrow \infty$. There are a number of ways to recognize convergence of the iterates. One way is to carry out a similarity transformation.

Let $\hat{Q}_m \in \mathbb{F}^{n \times n}$ denote the unitary matrix whose columns are $q_1^{(m)}, \dots, q_n^{(m)}$, and let

$$A_m = \hat{Q}_m^* A \hat{Q}_m. \quad (6.2.7)$$

For large m the space spanned by the first k columns of $\hat{Q}_m$ is very close to the invariant subspace $\mathcal{T}_k$. In Theorem 6.1.6 we learned that if these columns exactly span $\mathcal{T}_k$, then A_m has the block upper-triangular form

$$\begin{array}{cc} k & \begin{bmatrix} A_{11}^{(m)} & A_{12}^{(m)} \\ 0 & A_{22}^{(m)} \end{bmatrix} \\ n-k & \begin{array}{c} k \quad n-k \end{array} \end{array} \quad (6.2.8)$$

and the eigenvalues of $A_{11}^{(m)}$ are $\lambda_1, \dots, \lambda_k$. Since the columns do not exactly span $\mathcal{T}_k$, we do not get a block of zeros, but we have reason to hope that the entries in the block $A_{21}^{(m)}$ will be close to zero. It is not unreasonable to expect that $A_{21}^{(m)} \rightarrow 0$ at the same rate as $A^m S_k \rightarrow \mathcal{T}_k$. This turns out to be true and not hard to prove. See [79], for example. Thus the sequence (A_m) converges to the block upper-triangular form (6.2.8).

This happens not just for one choice of k , but for all k simultaneously, so the limiting form is upper triangular. The main-diagonal entries of A_m converge to the eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$, in order. These conclusions hold under the assumption that the strict inequalities $|\lambda_k| > |\lambda_{k+1}|$ hold for $k = 1, \dots, n-1$. Should some of them fail to hold, the limiting form will be block triangular. This happens most frequently when A is real and has some pairs of complex conjugate eigenvalues. For each pair $\lambda_i = \bar{\lambda}_{i+1}$, a 2-by-2 block will emerge in rows and columns i and $i+1$. The eigenvalues of this block are, of course, λ_i and λ_{i+1} .

The QR algorithm is a variant of subspace iteration that produces the sequence (A_m) directly.

The QR Algorithm

It is easy to recast simultaneous iteration in matrix form. After m iterations we have the orthonormal vectors $q_1^{(m)}, \dots, q_n^{(m)}$, which are the columns of $\hat{Q}_m$. Let B_{m+1} be the matrix whose columns are $Aq_1^{(m)}, \dots, Aq_n^{(m)}$. Then $B_{m+1} = A\hat{Q}_m$. To complete the step we must orthonormalize $Aq_1^{(m)}, \dots, Aq_n^{(m)}$. From Section 3.4 we know that this can be accomplished by a QR decomposition of B_{m+1} . Thus a step of simultaneous iteration can be expressed in matrix form as

$$A\hat{Q}_m = B_{m+1} = \hat{Q}_{m+1}R_{m+1}. \quad (6.2.9)$$

Exercise 6.2.10 Write down the matrix form of simultaneous iteration for the case when only k vectors are used ($k < n$). What are the dimensions and other properties of the matrices that are involved? $\square$

Now suppose we begin subspace iteration with the standard basis vectors $q_1^{(0)} = e_1, q_2^{(0)} = e_2, \dots, q_n^{(0)} = e_n$, that is, $\hat{Q}_0 = I$. Then by (6.2.9) we have $AI = B_1 = \hat{Q}_1 R_1$. Letting $Q_1 = \hat{Q}_1$, we have

$$A = Q_1 R_1. \quad (6.2.11)$$

Suppose we are feeling very optimistic about the rate of convergence, and after this one step we already wish to assess how much progress we have made. One way to do this is to carry out the unitary similarity transformation $A_1 = \hat{Q}_1^* A \hat{Q}_1 = Q_1^* A Q_1$, as suggested above (cf. (6.2.7)), and see how close to upper triangular form A_1 is. Since $Q_1^* A = R_1$ by (6.2.11), A_1 can be calculated by

$$R_1 Q_1 = A_1. \quad (6.2.12)$$

Equations (6.2.11) and (6.2.12) together constitute one step of the QR algorithm.

Finding that A_1 is not upper triangular, we decide to take another step. But now we have a choice. We can continue simultaneous iteration with A , or we can work instead with the similar matrix A_1 . Recall that similar matrices can be regarded as representations of the same linear transformation with respect to different coordinate systems. Thus the choice between A and A_1 is simply a choice between coordinate systems. If we stick with the original coordinates, the next step has the form

$$A \hat{Q}_1 = B_2 = \hat{Q}_2 R_2. \quad (6.2.13)$$

If we wish to assess our progress toward convergence, we can make the similarity transformation

$$A_2 = \hat{Q}_2^* A \hat{Q}_2. \quad (6.2.14)$$

Let us see what these transformations look like in the coordinate system of A_1 . Because $A_1 = \hat{Q}_1^* A \hat{Q}_1$, a vector that is represented by the coordinate vector x in the original system will be represented by $\hat{Q}_1^* x$ in the new system. Therefore the vectors $q_1^{(1)}, \dots, q_n^{(1)}$ (the columns of $\hat{Q}_1$) are transformed to $\hat{Q}_1^* q_1^{(1)} = e_1, \hat{Q}_1^* q_2^{(1)} = e_2, \dots, \hat{Q}_1^* q_n^{(1)} = e_n$ (the columns of I). Thus the equation $A \hat{Q}_1 = B_2$ is equivalent to $A_1 I = A_1$, and the QR decomposition $B_2 = \hat{Q}_2 R_2$ is equivalent to a QR decomposition of A_1 :

$$A_1 = Q_2 R_2. \quad (6.2.15)$$

If we now wish to see how much progress we have made in this one step, we can make the similarity transformation

$$A_2 = Q_2^* A_1 Q_2. \quad (6.2.16)$$

Since $R_2 = Q_2^* A$, this transformation can be accomplished via

$$R_2 Q_2 = A_2. \quad (6.2.17)$$

Equations (6.2.15) and (6.2.17) constitute a second QR step.

Exercise 6.2.18 Assume (for convenience) A is nonsingular. Therefore B_2 is also nonsingular, so its QR decomposition is unique (Theorem 3.2.58). One expression for the QR decomposition of B_2 is given by (6.2.13). Obtain a second expression by rewriting the equation $B_2 = A\hat{Q}_1$ in terms of A_1 and using (6.2.15). Conclude that the matrices R_2 given by (6.2.13) and (6.2.15) are the same, $\hat{Q}_2 = Q_1Q_2$, and the matrices A_2 given by (6.2.14) and (6.2.16) are the same. $\square$

When we continue the process, we have the choice of working in the coordinate system of A , A_1 , or A_2 . If we decide that at each step we will work in the newest coordinate system, we generate the sequence (A_m) by the elegant transformation

$$A_{m-1} = Q_m R_m \quad R_m Q_m = A_m. \quad (6.2.19)$$

This is the basic QR algorithm. As we have just seen, it is nothing but simultaneous iteration with a change of coordinate system at each step. To solidify your understanding of this fact, let us review the meaning of the equations in (6.2.19).

The columns of A_{m-1} are $A_{m-1}e_1, \dots, A_{m-1}e_n$; that is, they are the result of one step of simultaneous iteration by A_{m-1} on the standard basis. The decomposition $A_{m-1} = Q_m R_m$ orthonormalizes these vectors. The columns of Q_m are the orthonormal basis vectors for the next step of simultaneous iteration. The step $R_m Q_m = A_m$, which is equivalent to $A_m = Q_m^* A_{m-1} Q_m$, simply carries out the transformation to a new coordinate system. In this coordinate transformation, the orthonormal basis vectors for the next iteration are transformed to the standard basis vectors $e_1, \dots, e_n$.

The global correspondence between the QR algorithm and simultaneous iteration without coordinate transformations is easy to establish. The matrices A_m generated by (6.2.19) are the same as those generated by (6.2.7), assuming the iterations were initiated with $\hat{Q}_0 = I$. The R_m of (6.2.19) are the same as those of (6.2.9), and the Q_m of (6.2.19) are related to the $\hat{Q}_m$ of (6.2.9) by

$$\hat{Q}_m = Q_1 Q_2 \cdots Q_m. \quad (6.2.20)$$

Q_m is the matrix of the change of coordinates at step m , and $\hat{Q}_m$ is the accumulated change of coordinates after m steps.

We have established the equivalence of simultaneous iteration and the QR algorithm by looking at the processes one step at a time. Another way is to examine the cumulative effect of m steps. In this approach Q_m , R_m , and A_m are defined by (6.2.19), with $A_0 = A$, and $\hat{Q}_m$ is defined by (6.2.20). If, in addition, $\hat{R}_m$ is defined by

$$\hat{R}_m = R_m R_{m-1} \cdots R_1, \quad (6.2.21)$$

then

$$\begin{aligned} A &= Q_1 R_1 = \hat{Q}_1 \hat{R}_1, \\ A^2 &= Q_1 R_1 Q_1 R_1 = Q_1 Q_2 R_2 R_1 = \hat{Q}_2 \hat{R}_2, \\ A^3 &= Q_1 R_1 Q_1 R_1 Q_1 R_1 = Q_1 Q_2 R_2 Q_2 R_2 R_1 = Q_1 Q_2 Q_3 R_3 R_2 R_1 = \hat{Q}_3 \hat{R}_3, \end{aligned}$$

and so on. Clearly we can prove by induction that

$$A^m = \hat{Q}_m \hat{R}_m, \quad m = 1, 2, 3, \dots \quad (6.2.22)$$

Exercise 6.2.23 Suppose A_m , Q_m , and R_m are defined by (6.2.19), with $A_0 = A$, $\hat{Q}_m$ is defined by (6.2.20), and $\hat{R}_m$ is defined by (6.2.21).

- Prove that $A_m = \hat{Q}_m^* A \hat{Q}_m$, hence $A \hat{Q}_m = \hat{Q}_m A_m$, for all m .
- Prove (6.2.22) by induction on m , using the result from part (a).

□

Clearly $\hat{Q}_m$ is unitary and $\hat{R}_m$ is upper triangular. Therefore (6.2.22) is just the QR decomposition of A^m . This means that for all k , the first k columns of $\hat{Q}_m$ form an orthonormal basis for the space spanned by the first k columns of A^m . But the columns of A^m are just $A^m e_1, \dots, A^m e_n$. Thus

$$A^m \text{span}\{e_1, \dots, e_k\} = \text{span}\{q_1^{(m)}, \dots, q_k^{(m)}\}, \quad k = 1, \dots, n.$$

That is, the columns of $\hat{Q}_m$ are just the result of m steps of simultaneous iteration, starting from the standard basis vectors $e_1, \dots, e_n$.

If you decided to read this section before reading Section 5.6, now is a good time to go back and read that material. If you have already read Section 5.6, you might find it useful to review it at this point.

In Section 5.6 we introduced the QR algorithm without motivation and discussed the modifications that are needed to make the algorithm efficient. The first of these is to transform the matrix to upper Hessenberg form before beginning the QR iterations. This is a good idea because (i) Hessenberg form is preserved under QR iterations and (ii) the cost of a QR iteration is much less for a Hessenberg matrix than for a full matrix. Thus each of our iterates has the form

$$A_m = \begin{bmatrix} a_{11}^{(m)} & a_{12}^{(m)} & \cdots & a_{1,n-1}^{(m)} & a_{1n}^{(m)} \\ a_{21}^{(m)} & a_{22}^{(m)} & \cdots & a_{2,n-1}^{(m)} & a_{2n}^{(m)} \\ & a_{32}^{(m)} & \cdots & a_{3,n-1}^{(m)} & a_{3n}^{(m)} \\ & & \ddots & \vdots & \vdots \\ & & & a_{n,n-1}^{(m)} & a_{nn}^{(m)} \end{bmatrix}. \quad (6.2.24)$$

We may assume further that each of the subdiagonal entries $a_{k+1,k}^{(m)}$ is nonzero, since otherwise we could reduce the eigenvalue problem to two smaller problems. Recall that an upper Hessenberg matrix that satisfies this condition is called a properly upper Hessenberg matrix.

Since the matrices A_m produced by the QR algorithm are identical to those given by (6.2.7), we can conclude that if $|\lambda_k| > |\lambda_{k+1}|$ and the subspace condition $\mathcal{S}_k \cap \mathcal{U}_k = \{0\}$ (Theorem 6.2.3) holds, then A_m converges to the block triangular form (6.2.8). Since all A_m are upper Hessenberg, the $(n-k) \times k$ block that is converging

to zero contains only one entry that is not already zero, namely $a_{k+1,k}^{(m)}$. This one entry therefore gives a rough indication of the distance of $\text{span}\{q_1^{(m)}, \dots, q_k^{(m)}\}$ from the invariant subspace $\mathcal{T}_k = \text{span}\{v_1, \dots, v_k\}$. Under the given conditions $a_{k+1,k}^{(m)} \rightarrow 0$ linearly with convergence ratio $|\lambda_{k+1}/\lambda_k|$ as $m \rightarrow \infty$.

A desirable side effect of using properly upper Hessenberg matrices is that the subspace conditions $\mathcal{S}_k \cap \mathcal{U}_k = \{0\}$, $k = 1, \dots, n-1$, are always satisfied (Exercise 6.2.34). This is a very satisfying result from a theoretical standpoint. It implies that the eigenvalue inequalities $|\lambda_k| > |\lambda_{k+1}|$ are all that is needed to guarantee convergence.

A word of caution about the nature of the convergence is in order. The statement that (A_m) converges to upper-triangular form means simply that the subdiagonal entries $a_{k+1,k}^{(m)}$ converge to zero as $m \rightarrow \infty$. It follows that the main-diagonal entries converge to eigenvalues, but nothing is implied about the entries above the main diagonal, which may in fact fail to converge (Exercise 6.2.35). Thus we *cannot* assert that there is some upper-triangular $U \in \mathbb{F}^{n \times n}$ such that $\|A_m - U\| \rightarrow 0$ as $m \rightarrow \infty$.

Along with the use of upper Hessenberg matrices, the other major modification of the QR algorithm is the use of shifts to accelerate convergence. Now that we have established that the rate of convergence is determined by the ratios $|\lambda_{k+1}/\lambda_k|$, it is clear how shifts can be used. This was discussed in Section 5.6, and there is no need to repeat that discussion here. Perhaps one point does bear repeating. While it is clear that shifts are normally extremely beneficial, their use does complicate the convergence analysis. Nobody has been able to prove that the QR algorithm (with some specific, evidently successful shifting strategy) always converges.

The generalization of (6.2.22) that holds for the shifted QR algorithm is developed in Exercise 6.2.36.

Duality in Subspace Iteration

The following duality theorem provides the link between the QR algorithm and inverse iteration. It shows that whenever direct (subspace) iteration takes place, inverse (subspace) iteration also takes place automatically.

Theorem 6.2.25 *Let $A \in \mathbb{F}^{n \times n}$ be nonsingular, and let $B = (A^*)^{-1}$. Let S be any subspace of $\mathbb{F}^n$. Then the two sequences of subspaces*

$$\begin{aligned} S, AS, A^2S, \dots \\ S^\perp, BS^\perp, B^2S^\perp, \dots \end{aligned}$$

are equivalent in the sense that they yield orthogonal complements. That is,

$$(A^m S)^\perp = B^m (S^\perp) \quad \text{for } m = 0, 1, 2, \dots$$

Proof. For every $x, y \in \mathbb{F}^n$, $\langle A^m x, B^m y \rangle = y^* (B^*)^m A^m x = y^* (A^{-1})^m A^m x = y^* x = \langle x, y \rangle$. Thus $A^m x$ and $B^m y$ are orthogonal if and only if x and y are orthogonal. The theorem follows directly from this observation. $\square$

The second subspace sequence in Theorem 6.2.25 is a sequence of inverse iterates, since the iteration matrix is the inverse of A^* .

Let us first see how Theorem 6.2.25 applies to the unshifted QR algorithm, which is essentially simultaneous iteration with starting subspaces $\text{span}\{e_1, \dots, e_k\}$, $k = 1, \dots, n$. After m steps we have

$$A^m \text{span}\{e_1, \dots, e_k\} = \text{span}\{q_1^{(m)}, \dots, q_k^{(m)}\}, \quad k = 1, \dots, n-1, \quad (6.2.26)$$

where $q_1^{(m)}, \dots, q_n^{(m)}$ are the columns of $\hat{Q}_m$. Since

$$\text{span}\{e_1, \dots, e_k\}^\perp = \text{span}\{e_{k+1}, \dots, e_n\}$$

and $\text{span}\{q_1^{(m)}, \dots, q_k^{(m)}\}^\perp = \text{span}\{q_{k+1}^{(m)}, \dots, q_n^{(m)}\}$, Theorem 6.2.25 applied to (6.2.26) yields

$$(A^*)^{-m} \text{span}\{e_{k+1}, \dots, e_n\} = \text{span}\{q_{k+1}^{(m)}, \dots, q_n^{(m)}\}, \quad k = 1, \dots, n-1. \quad (6.2.27)$$

Of particular interest is the case $k = n-1$:

$$(A^*)^{-m} \text{span}\{e_n\} = \text{span}\{q_n^{(m)}\}.$$

Thus the last column of $\hat{Q}_m$ is seen to be the result of single-vector inverse iteration on A^* . It should therefore be possible to cause this column to converge rapidly by applying shifts to A^* . Before we investigate this possibility, let us take time to see how (6.2.27) can be derived directly from the basic equations of the QR algorithm. This will reveal some interesting aspects of duality.

The QR sequence (A_m) is generated by (6.2.19) with $A_0 = A$. Let $B_m = (A_m^*)^{-1}$ and $L_m = (R_m^*)^{-1}$. Note that L_m is lower triangular with positive entries on the main diagonal. Take conjugate transposes and inverses in (6.2.19) to obtain

$$B_{m-1} = Q_m L_m \quad L_m Q_m = B_m \quad m = 1, 2, 3, \dots$$

with $B_0 = B = (A^*)^{-1}$. This shows that the QR algorithm applied to A is equivalent to a QL algorithm applied to $(A^*)^{-1}$. The theory and algorithms based on the QL decomposition are essentially identical to those based on QR . The next two exercises develop some basic facts about the QL decomposition.

Exercise 6.2.28 Let $B \in \mathbb{F}^{n \times n}$ be nonsingular. Prove that there exist unique matrices $Q, L \in \mathbb{F}^{n \times n}$ such that Q is unitary, L is lower triangular with positive entries on the main diagonal, and $B = QL$. (You may prove this from scratch or deduce it from the corresponding theorem about QR decompositions.) $\square$

Exercise 6.2.29 Let $B = QL$ with $B = [b_1 \cdots b_n]$ nonsingular, $Q = [q_1 \cdots q_n]$ unitary, and L lower triangular. Show that $\text{span}\{q_n\} = \text{span}\{b_n\}$, $\text{span}\{q_{n-1}, q_n\} = \text{span}\{b_{n-1}, b_n\}$, and in general

$$\text{span}\{q_{k+1}, \dots, q_n\} = \text{span}\{b_{k+1}, \dots, b_n\} \quad k = n-1, n-2, \dots, 0.$$

□

Exercise 6.2.29 shows that the QL decomposition is equivalent to a Gram-Schmidt procedure that orthonormalizes the columns of B from right to left.

Recalling that $\hat{Q}_m = Q_1 \cdots Q_m$ and $\hat{R}_m = R_m \cdots R_1$, define $\hat{L}_m = (\hat{R}_m^*)^{-1} = L_m \cdots L_1$. Clearly $\hat{L}_m$ is lower triangular with positive main-diagonal entries. Taking conjugate transposes and inverses in (6.2.22), we obtain

$$B_m = \hat{Q}_m \hat{L}_m \quad m = 1, 2, 3, \dots$$

Applying the conclusion of Exercise 6.2.29 to this QL decomposition, we find that

$$B^m \text{span}\{e_{k+1}, \dots, e_n\} = \text{span}\{q_{k+1}^m, \dots, q_n^{(m)}\}, \quad k = 1, \dots, n-1.$$

This is just (6.2.27) again.

Now let us apply duality to the study of shifts of origin. Specifically, we will demonstrate that the QR algorithm with Rayleigh quotient shifts is, in part, Rayleigh quotient iteration. We can accomplish this by examining the cumulative effect of m steps (Exercise 6.2.39) or by taking a step-at-a-time point of view. The latter is probably more revealing, so we will take that approach.

Consider a single QR step

$$A_{m-1} - \rho_{m-1}I = Q_m R_m \quad R_m Q_m + \rho_{m-1}I = A_m,$$

where $\rho_{m-1} = a_{nn}^{(m-1)} = e_n^* A_{m-1} e_n$. This is the Rayleigh quotient shift defined in Section 5.6. Then $\bar{\rho}_{m-1}$ is a Rayleigh quotient of A_m^* , since $\bar{\rho}_{m-1} = e_n^* A_m^* e_n$. If $|a_{n,n-1}|^{(m-1)}$ is small enough, then $\bar{\rho}_{m-1}$ is a good approximation to an eigenvalue of A_{m-1}^* . It is also easy to see, at least informally, that e_n is approximately an eigenvector of A_{m-1}^* (cf. Exercise 5.6.26). Let $p_1, \dots, p_n$ denote the columns of Q_m . From the equation $A_{m-1} - \rho_{m-1}I = Q_m R_m$, it follows that

$$(A_{m-1} - \rho_{m-1}I) \text{span}\{e_1, \dots, e_{n-1}\} = \text{span}\{p_1, \dots, p_{n-1}\}.$$

Applying Theorem 6.2.25 to this equation, we obtain

$$(A_{m-1}^* - \bar{\rho}_{m-1}I)^{-1} \text{span}\{e_n\} = \text{span}\{p_n\}. \quad (6.2.30)$$

Since $\bar{\rho}_{m-1}$ is the Rayleigh quotient formed from A_{m-1}^* and e_n , the equation (6.2.30) represents a step of Rayleigh quotient iteration. The result is p_n , the last column of Q_m .

Now suppose we were to take another step of Rayleigh quotient iteration using A_{m-1}^* . The shift would be $\bar{\rho} = p_n^* A_{m-1}^* p_n$, and we would calculate

$$(A_{m-1}^* - \bar{\rho}I)^{-1} p_n. \quad (6.2.31)$$

Using the equation $A_m = Q_m^* A_{m-1} Q_m$, we see easily that $\rho = a_{nn}^{(m)}$. If, on the other hand, we take another QR step, we will operate on the matrix A_m and choose the

shift $\rho = e_n^* A_m e_n = a_{nn}^{(m)}$, which is the same as ρ . This is hardly a coincidence. The coordinate transformation that maps A_{m-1} to A_m also transforms p_n to e_n , so the computations $\rho = p_n^* A_{m-1} p_n$ and $\rho_n = e_n^* A_m e_n$ represent the same computation in two different coordinate systems. By (6.2.30) with $m - 1$ replaced by m , the QR step on A_m calculates (among other things)

$$(A_m^* - \bar{\rho}_m I)^{-1} e_n.$$

This computation is the same as (6.2.31), except that the coordinate system has been changed. This proves the QR algorithm is, in part, Rayleigh quotient iteration. As a consequence, $a_{nn}^{(m)}$ (normally) converges to an eigenvalue quadratically, as that is the usual convergence rate of Rayleigh quotient iteration. At the same time $\text{span}\{q_n^{(m)}\}$, the last column of $\hat{Q}_m$, (normally) converges quadratically to an eigenvector of A^* (cf. Exercise 6.2.39). If A is Hermitian, or even just normal, the convergence is cubic.

Exercise 6.2.32 Verify that $Q_m^* p_n = e_n$ and $Q_m^* [(A_{m-1}^* - \bar{\rho}_m I)^{-1} p_n] = (A_m^* - \bar{\rho}_m I)^{-1} e_n$. $\square$

Additional Exercises

Exercise 6.2.33 Let S and U be subspaces of $\mathbb{F}^n$ of dimension k and $n - k$, with bases $s_1, \dots, s_k$ and $u_1, \dots, u_{n-k}$, respectively. Let $B = [s_1 \cdots s_k \ u_1 \cdots u_{n-k}] \in \mathbb{F}^{n \times n}$. Show that $S \cap U = \{0\}$ if and only if B is nonsingular. (B fails to be nonsingular when and only when $\det(B) = 0$. This is a very special relationship. If one chooses n vectors at random from $\mathbb{F}^n$ and builds a matrix whose columns are those vectors, its determinant is almost certain to be nonzero.) $\square$

Exercise 6.2.34 The QR algorithm is essentially simultaneous (subspace) iteration with starting subspaces $S_k = \text{span}\{e_1, \dots, e_k\}$, $k = 1, \dots, n - 1$. Theorem 6.2.3 states that subspace iteration converges if $|\lambda_k| > |\lambda_{k+1}|$ and the subspace condition $S_k \cap U_k = \{0\}$ holds. Recall that $U_k = \text{span}\{v_{k+1}, \dots, v_n\}$, which is a subspace of $\mathbb{F}^n$ of dimension $n - k$ that is invariant under A . In this exercise you will show that the subspace condition $S_k \cap U_k = \{0\}$ is automatically satisfied if A is properly upper Hessenberg.

- Let $\mathcal{V}$ be a subspace of $\mathbb{F}^n$ that is invariant under A . Show that if $v \in \mathcal{V}$, then $A^m v \in \mathcal{V}$ for $m = 1, 2, 3, \dots$
- Suppose A is properly upper Hessenberg. Let $v \in \text{span}\{e_1, \dots, e_k\} = S_k$ be nonzero. (What does v look like? What can you say about Av ? $A^2 v$?) Show that the vectors $v, Av, A^2 v, \dots, A^{n-k} v$ are linearly independent.
- Show that v cannot lie in U_k . Hence $S_k \cap U_k = \{0\}$.

$\square$

Exercise 6.2.35 Normally one would have no interest in applying the QR algorithm to the matrix

$$A = \begin{bmatrix} 2 & 1 \\ 0 & -1 \end{bmatrix},$$

since it is already upper triangular. However, it is a good matrix to illustrate a simple point about convergence of the QR algorithm. Calculate the QR iterates $A = A_0, A_1, A_2, \dots$. In each QR decomposition choose R_m so that its main-diagonal entries are all positive. Note that $a_{12}^{(m)}$ does not converge as $m \rightarrow \infty$. $\square$

Exercise 6.2.36 Let (A_m) be generated by the shifted QR algorithm:

$$A_{m-1} - \rho_{m-1}I = Q_m R_m, \quad R_m Q_m + \rho_{m-1}I = A_m, \quad m = 1, 2, 3, \dots,$$

where $A_0 = A$. Let $\hat{Q}_m = Q_1 \cdots Q_m$ and $\hat{R}_m = R_m \cdots R_1$.

- Show that $A_m = \hat{Q}_m^* A \hat{Q}_m$ for $m = 1, 2, 3, \dots$
- Deduce that $(A - \rho_m I) \hat{Q}_m = \hat{Q}_m (A_m - \rho_m I)$ for $m = 1, 2, 3, \dots$
- Now prove by induction on m that

$$(A - \rho_{m-1}I) \cdots (A - \rho_0 I) = \hat{Q}_m \hat{R}_m, \quad m = 1, 2, 3, \dots \quad (6.2.37)$$

This equation generalizes (6.2.22). Notice that it is also a generalization of Lemma 5.7.16, which is one of the crucial results in the development of the double-step QR algorithm. $\square$

Exercise 6.2.38 We continue the discussion of the shifted QR algorithm from the previous exercise.

- Show that if ρ_{m-1} is not an eigenvalue of A , then R_m is nonsingular, and $A_m = R_m A_{m-1} R_m^{-1}$.
- Show that if none of $\rho_0, \dots, \rho_{m-1}$ is an eigenvalue of A , then $\hat{R}_m$ is nonsingular, and $A_m = \hat{R}_m A \hat{R}_m^{-1}$.

$\square$

Exercise 6.2.39 This exercise shows, by the cumulative viewpoint, that the QR algorithm effects Rayleigh quotient iteration. If m steps of the QR algorithm with Rayleigh quotient shift are performed, then (6.2.37) holds, where $\rho_i = a_{nn}^{(i)}$ for $i = 0, \dots, m-1$. Apply Theorem 6.2.25 to (6.2.37), or take conjugate transpose inverses, to obtain information about $\text{span}\{q_n^{(m)}\}$. Show that $\text{span}\{q_n^{(m)}\}$ is exactly the space obtained after m steps of Rayleigh quotient iteration applied to A^* with starting vector e_n . In particular, be sure to check that each shift is the appropriate Rayleigh quotient for that step. $\square$

6.3 EIGENVALUES OF LARGE, SPARSE MATRICES, I

Most large matrices that occur in applications are sparse. That is, the vast majority of their entries are zeros. If a matrix is sparse enough, it may be worthwhile to store it in a sparse data structure, which stores only the nonzero entries, together with information about where each entry belongs in the matrix. If the matrix is extremely large, there may be no alternative to the sparse data structure, since there may not be enough storage space to store the matrix in the conventional way. If we want to find some eigenvalues of such a matrix, we need to use a method that can work with the sparse data structure. The QR algorithm and other algorithms that use similarity transformations are inappropriate here, as they cause a great deal of fill-in: after one QR iteration, the resulting matrix A_1 is hardly sparse at all (see Exercise 6.3.24).

Needed are methods that do not alter the matrix. One algorithm that immediately comes to mind is simultaneous iteration (6.2.6). Here we refer to the basic simultaneous iteration algorithm, which does not change the coordinate system at each step. Looking at (6.2.6), we see that the only way the algorithm uses A is to multiply it by the vectors $q_1^{(m)}, \dots, q_k^{(m)}$ at each step. The entries of A are never altered in any way. Even if A is stored in a sparse format, it is a simple matter to calculate a matrix-vector product Aq (see Exercise 6.3.25). Thus it is a simple matter to perform simultaneous iteration on a large, sparse matrix. The number of vectors, k , that we use in the simultaneous iteration is typically limited by storage space and computational requirements, so we normally have $k \ll n$, which means that we can compute at most the few (at most k) eigenvalues of largest modulus and the associated eigenvectors. Some of the practical details of simultaneous iteration are studied in the following exercise.

Exercise 6.3.1 Consider the simultaneous iteration algorithm (6.2.6) with $k \ll n$.

- (a) Let $q_1^{(m)}, \dots, q_k^{(m)}$ be the k orthonormal vectors at the m th iteration of simultaneous iteration (6.2.6), and let $\hat{Q}_m = [q_1^{(m)} \cdots q_k^{(m)}] \in \mathbb{R}^{n \times k}$. This is a tall, skinny matrix. Show that

$$A\hat{Q}_m = \hat{Q}_{m+1}R_{m+1},$$

where R_{m+1} is a $k \times k$ upper-triangular matrix (cf. (6.2.9) and Exercise 6.2.10). This is a condensed QR decomposition.

- (b) Let $B_m = A\hat{Q}_m \in \mathbb{C}^{n \times k}$ and $C_m = \hat{Q}_m^* B_m \in \mathbb{C}^{k \times k}$. Show that the columns of $\hat{Q}_m$ span an invariant subspace under A if and only if $B_m - \hat{Q}_m C_m = 0$, in which case the eigenvalues of the small matrix C_m are k of the eigenvalues of A . (In practice we can accept $\hat{Q}_m$ as invariant as soon as $\|B_m - \hat{Q}_m C_m\|_1$ is sufficiently small.)
- (c) Show how invariant subspaces of dimension $j = 1, 2, \dots, k-1$ can be detected by considering leading submatrices of $\hat{Q}_m$, B_m , and C_m . Once an invariant subspace of dimension j has been detected, the first j columns of $\hat{Q}_m$

can be “locked in,” and subsequent iterations can operate on the subset $q_{j+1}^{(m)}, \dots, q_k^{(m)}$.

- (d) A step of simultaneous iteration has three stages: (i) computation of B_m , (ii) condensed QR decomposition of B_m , and (iii) tests for convergence of invariant subspaces. Estimate the number of flops that each of these stages requires (See Exercise 6.3.25). About how much storage space is needed in all? In typical large applications k is small, and the first stage dominates the computational cost. Show, however, that if k is made large enough, the other stages will dominate the flop count.

□

Up until about 1980, simultaneous iteration was the most popular method for computing a few eigenvalues of a large, sparse matrix. A number of refinements and acceleration techniques were introduced, but ultimately simultaneous iteration was supplanted by more sophisticated methods, some of which we will describe in this section.

The Shift-and-Invert Strategy

Before we leave simultaneous iteration, we will describe the *shift-and-invert* strategy, which can be combined with simultaneous iteration to compute interior eigenvalues of A . As we shall later see, the shift-and-invert strategy can also be used in conjunction with other algorithms.

Simultaneous iteration computes the eigenvalues of A of largest modulus. What if that's not what we want? Suppose we would like to compute the eigenvalues in a certain region near some target value τ . If we shift by τ , and then take the inverse, we get a new matrix $(A - \tau I)^{-1}$ that has the same eigenvectors and invariant subspaces as A but different eigenvalues. Each eigenvalue λ of A corresponds to an eigenvalue $(\lambda - \tau)^{-1}$ of $(A - \tau I)^{-1}$. The largest eigenvalues of $(A - \tau I)^{-1}$ correspond to the eigenvalues of A that are closest to τ . We can find these eigenvalues (and associated invariant subspaces) by applying simultaneous iteration to $(A - \tau I)^{-1}$. This is the shift-and-invert strategy. Of course we don't actually form the inverse matrix; the inverse of a sparse matrix is not sparse. Instead we compute (once!) a sparse LU decomposition of $A - \tau I$. The computation $(A - \tau I)^{-1}q = p$ is then effected by solving for p the equation $(A - \tau I)p = q$ by forward and back substitution, using the factors of the LU decomposition. Once we have the LU decomposition in hand, we can use it over and over again to perform as many computations of type $(A - \tau I)^{-1}q$ as we please. A new LU decomposition is not needed unless we change the target shift τ .

The shift-and-invert technique is quite powerful and has found widespread use over the years. However, there is an important limitation on its applicability. If A is really large, its sparse LU decomposition, which is typically much less sparse than A itself, will require a great deal of storage space. If the computer's memory is not big enough to store the LU decomposition, then we cannot use this technique.

The Arnoldi Process

The Arnoldi process [2] has been around since about 1950, but it did not come into vogue as a method for computing eigenvalues until the 1970's. Variants of the Arnoldi process have now supplanted simultaneous iteration as the method of choice for a wide variety of applications.

Recall that the power method starts with an initial vector q and then computes (multiples of) the vectors $q, Aq, A^2q, \dots, A^kq, \dots$. As the algorithm proceeds, it discards the old iterates. At step k , it has only a multiple of A^kq ; none of the past information has been saved. Simultaneous iteration operates on subspaces, but it works the same way; the information from the past is discarded.

The idea of the Arnoldi process is to retain and use all of the past information. After k steps we have the $k + 1$ vectors $q, Aq, \dots, A^kq$, all of which we have saved. We then search for good eigenvector approximations in the $(k + 1)$ -dimensional space spanned by these vectors. Now we consider some practical matters.

In practice the vectors $q, Aq, \dots, A^kq$ usually are an ill-conditioned basis for the space that they span. The vector $A^j q$ points more and more in the direction of a dominant eigenvector for A as j is increased. Thus the vectors toward the end of the sequence $q, \dots, A^kq$ may be pointing in nearly the same direction. To counter this we can replace these vectors by an orthonormal set $q_1, \dots, q_{k+1}$ that spans the same space. This can be achieved by the Gram-Schmidt process with one slight modification. Suppose we have generated the first k vectors and now we want to obtain the $(k + 1)$ st. If we were working with the original sequence $q, Aq, \dots, A^{k-1}q$, we would simply multiply $A^{k-1}q$ by A to obtain A^kq . However, we do not have the original sequence; we have the orthonormal set $q_1, \dots, q_k$, and we want q_{k+1} . In principle we can obtain q_{k+1} by orthogonalizing A^kq against $q_1, \dots, q_k$ using Gram-Schmidt. In practice this is not possible, because we do not have the vector A^kq . Thus we must do something else. Instead of multiplying the (unavailable) vector $A^{k-1}q$ by A to obtain A^kq , we multiply the (available) vector q_k by A to obtain Aq_k . We then orthogonalize Aq_k against $q_1, \dots, q_k$ to obtain q_{k+1} . This is the Arnoldi process.

Now let's add a bit more detail. The first step of the Arnoldi process consists of the normalization

$$q_1 = q / \|q\|_2. \quad (6.3.2)$$

On subsequent steps we take

$$\tilde{q}_{k+1} = Aq_k - \sum_{j=1}^k q_j h_{jk}, \quad (6.3.3)$$

where h_{jk} is the Gram-Schmidt coefficient

$$h_{jk} = \langle Aq_k, q_j \rangle. \quad (6.3.4)$$

This is just the right coefficient to make $\tilde{q}_{k+1}$ orthogonal to q_j . Thus $\tilde{q}_{k+1}$ is orthogonal to $q_1, \dots, q_k$ but does not have norm 1. We complete the step by

normalizing $\tilde{q}_{k+1}$. That is, we take

$$q_{k+1} = \tilde{q}_{k+1}/h_{k+1,k}, \quad \text{where} \quad h_{k+1,k} = \|\tilde{q}_{k+1}\|_2. \quad (6.3.5)$$

It is not hard to show that (in exact arithmetic) this process produces exactly the same sequence of vectors as the Gram-Schmidt process applied to $q, Aq, A^2q, \dots, A^k q$. This is partly demonstrated in Exercise 6.3.30.

Equation (6.3.4) is the theoretical Gram-Schmidt formula for h_{jk} . In practice we can compute h_{jk} by either the classical or the modified Gram-Schmidt process. To ensure orthogonality of the vectors, we may choose to use reorthogonalization. See the discussion in Section 3.4. The following algorithm does modified Gram-Schmidt with reorthogonalization.

Arnoldi Process. Given a nonzero starting vector q , this algorithm produces orthonormal $q_1, q_2, \dots, q_m$ such that

$$\text{span}\{q_1, \dots, q_k\} = \text{span}\{q, Aq, \dots, A^{k-1}q\} \quad \text{for } k = 1, \dots, m.$$

$$\begin{aligned} & q_1 = q/\|q\|_2 \\ & \text{for } k = 1, \dots, m-1 \\ & \quad \left[\begin{array}{l} q_{k+1} \leftarrow Aq_k \\ \text{for } j = 1, \dots, k \text{ (orthogonalize)} \\ \quad \left[\begin{array}{l} h_{jk} \leftarrow q_j^* q_{k+1} \\ q_{k+1} \leftarrow q_{k+1} - q_j h_{jk} \end{array} \right] \\ \text{for } j = 1, \dots, k \text{ (reorthogonalize)} \\ \quad \left[\begin{array}{l} \alpha \leftarrow q_j^* q_{k+1} \\ q_{k+1} \leftarrow q_{k+1} - q_j \alpha \\ h_{jk} \leftarrow h_{jk} + \alpha \end{array} \right] \\ h_{k+1,k} \leftarrow \|q_{k+1}\|_2 \\ \text{if } h_{k+1,k} = 0 \\ \quad \left[\begin{array}{l} \text{set flag (span}\{q_1, \dots, q_k\} \text{ is invariant under } A) \\ \text{exit} \end{array} \right] \\ q_{k+1} \leftarrow q_{k+1}/h_{k+1,k} \end{array} \right. \quad (6.3.6) \end{aligned}$$

In this algorithm the only reference to the matrix is in the line $q_{k+1} \leftarrow Aq_k$. Thus we can apply the algorithm to A as long as we can multiply A by an arbitrary vector.

Before we can show how to use the Arnoldi process to compute eigenvalues, we need to establish a few fundamental relationships. We begin with some notation and terminology. For any j , the space $\text{span}\{q, Aq, A^2q, \dots, A^{j-1}q\}$ is called the j th *Krylov subspace* associated with A and q and denoted $\mathcal{K}_j(A, q)$.

Proposition 6.3.7 Suppose $q, Aq, \dots, A^{m-1}q$ are linearly independent. Then $\mathcal{K}_m(A, q)$ is invariant under A if and only if $q, Aq, \dots, A^{m-1}q, A^m q$ are linearly dependent.

Exercise 6.3.8 Prove Proposition 6.3.7. □

Theorem 6.3.9 Suppose $q, Aq, \dots, A^{m-1}q$ are linearly independent, and $q_1, \dots, q_m$ are generated by the Arnoldi process (6.3.2), (6.3.3), (6.3.4), (6.3.5). Then

- (a) $\text{span}\{q_1, \dots, q_k\} = \mathcal{K}_k(A, q)$ for $k = 1, \dots, m$.
- (b) For $k = 1, \dots, m-1$, $h_{k+1,k} > 0$.
- (c) $h_{m+1,m} = 0$ if and only if $q, Aq, \dots, A^m q$ are linearly dependent, which holds in turn if and only if the Krylov subspace $\mathcal{K}_m(A, q)$ is invariant under A . This justifies the flag in Algorithm 6.3.6.

Theorem 6.3.9 can be proved by induction on k . See Exercise 6.3.30. If you are not planning to work that exercise right away, you can at least work the following special case.

Exercise 6.3.10 Prove Theorem 6.3.9 in the case $k = 2$. (The case $k = 1$ is trivial.)

- (a) Specifically, show that if q and Aq are linearly independent, then $\tilde{q}_2$ in (6.3.3) is nonzero. Thus $h_{21} > 0$, and the division in (6.3.5) can be performed to obtain q_2 . Show that there exist constants c_0 and c_1 , with $c_1 \neq 0$, such that $q_2 = c_0 q + c_1 Aq$. Show further that $q_1 \in \mathcal{K}_2(A, q)$, $q_2 \in \mathcal{K}_2(A, q)$, and hence $\text{span}\{q_1, q_2\} \subseteq \mathcal{K}_2(A, q)$. Conversely, show that $q \in \text{span}\{q_1, q_2\}$, $Aq \in \text{span}\{q_1, q_2\}$, and hence $\mathcal{K}_2(A, q) \subseteq \text{span}\{q_1, q_2\}$.
- (b) Show that if q and Aq are linearly dependent, then $\tilde{q}_2 = 0$, $h_{21} = 0$, and $\text{span}\{q\}$ is invariant under A . In this case q is an eigenvector of A .

□

Matrix Representations of the Arnoldi Process

From (6.3.3) and (6.3.5) we easily deduce that

$$Aq_k = \sum_{j=1}^{k+1} q_j h_{jk}, \quad k = 1, 2, 3, \dots \quad (6.3.11)$$

From Theorem 6.3.9 we know that these relationships hold for $k = 1, \dots, m$ if $q, Aq, \dots, A^m q$ are linearly independent. These $m+1$ vector equations can be combined into a single matrix equation as follows. Define

$$Q_m = [q_1 \quad \cdots \quad q_m] \in \mathbb{C}^{n \times m}$$

and

$$H_{m+1,m} = \begin{bmatrix} h_{11} & h_{12} & \cdots & h_{1,m-1} & h_{1m} \\ h_{21} & h_{22} & \cdots & h_{2,m-1} & h_{2m} \\ 0 & h_{32} & \cdots & h_{3,m-1} & h_{3m} \\ \vdots & & \ddots & & \vdots \\ 0 & & & h_{m,m-1} & h_{mm} \\ 0 & 0 & \cdots & 0 & h_{m+1,m} \end{bmatrix} \in \mathbb{C}^{(m+1) \times m}.$$

Since Q_m has orthonormal columns, it is an isometry. $H_{m+1,m}$ is a non-square upper-Hessenberg matrix with strictly positive entries on the subdiagonal. From (6.3.11) we immediately have

$$AQ_m = Q_{m+1}H_{m+1,m}, \quad (6.3.12)$$

for the k th column of (6.3.12) is exactly (6.3.11). This matrix equation summarizes neatly the relationships between A and the quantities that are generated by the Arnoldi process. Another useful expression is obtained by rewriting the right-hand side of (6.3.12). Let H_m denote the square upper Hessenberg matrix obtained by deleting the bottom row of $H_{m+1,m}$. If we separate the last column of Q_{m+1} and the bottom row of $H_{m+1,m}$ from the matrix product, we obtain $Q_{m+1}H_{m+1,m} = Q_m H_m + q_{m+1} [0, \dots, 0, h_{m+1,m}]$. Thus

$$AQ_m = Q_m H_m + q_{m+1} h_{m+1,m} e_m^T, \quad (6.3.13)$$

where e_m denotes the m th standard basis vector in $\mathbb{R}^m$.

Exercise 6.3.14 Verify equations (6.3.11), (6.3.12), and (6.3.13). □

The following simple result, which is proved in Exercise 6.3.33, will be used in the next section.

Proposition 6.3.15 Suppose $q_1, \dots, q_{m+1}$ are orthonormal vectors,

$$Q_m = [q_1 \quad \dots \quad q_m],$$

and H_m is an upper Hessenberg matrix with $h_{j+1,j} > 0$ for $j = 1, \dots, m$. Although these may have been obtained by any means whatsoever, suppose they satisfy (6.3.13). Then $q_1, \dots, q_{m+1}$ must be exactly the vectors produced by the Arnoldi process with starting vector q_1 . In other words, given a matrix A , the objects in (6.3.13) are uniquely determined by the first column of Q_m .

If $q, Aq, \dots, A^m q$ are linearly independent, then $h_{m+1,m} \neq 0$. However, if they are dependent, we have $h_{m+1,m} = 0$, and (6.3.13) becomes $AQ_m = Q_m H_m$. We then conclude from Theorem 6.1.3 that the space spanned by the columns of Q_m is invariant under A . (This is consistent with Theorem 6.3.9, since $\mathcal{R}(Q_m) = \text{span}\{q_1, \dots, q_m\} = \mathcal{K}_m(A, q)$.) Furthermore, by Exercise 6.1.4 the eigenvalues of H_m are eigenvalues of A . If m is not too big, we can easily compute these m eigenvalues using, say, the QR algorithm. This begins to show how the Arnoldi process can deliver eigenvalues.

Our plan is to apply the Arnoldi process to find a few eigenvalues of extremely large matrices. Each step of the process adds another basis vector q_k , which takes up significant storage space. We also need to store the entries h_{jk} of the Hessenberg matrix. These storage requirements severely limit the number of steps we will be able to take. Thus we will typically have to stop after m steps, where $m \ll n$.

Nevertheless, there is nothing to stop us from imagining what happens if we take more steps. If $q, Aq, \dots, A^{n-1}q$ are independent, we can take n steps and end up

with $q_1, \dots, q_n$, an orthonormal basis of $\mathbb{C}^n$. If we then try to take one more step, we will get $h_{n+1,n} = 0$, since the $n+1$ vectors $q, Aq, \dots, A^n q$ in the n -dimensional space $\mathbb{C}^n$ must be linearly dependent. Thus (6.3.13) becomes $AQ_n = Q_n H_n$. The square matrix Q_n is unitary, H_n is properly upper Hessenberg, and we have $H_n = Q_n^{-1} A Q_n$. Thus the Arnoldi process, if carried to completion, computes a unitary similarity transformation to upper Hessenberg form. In Section 5.5 you learned how to accomplish this same task using reflectors. The process developed in Section 5.5 is generally preferred over the Arnoldi process for this particular task. See Exercise 6.3.34.

Now let us return to our main task, which is to obtain information about the eigenvalues of A by taking only a few steps of the Arnoldi process. We have already seen that if we get $h_{m+1,m} = 0$ at some point, then $\text{span}\{q_1, \dots, q_m\}$ is invariant under A , and the m eigenvalues of H_m are eigenvalues of A . However, we will seldom be so lucky as to encounter a small invariant subspace, unless we choose q very carefully. We'll find out how to do this by the implicitly restarted Arnoldi process in the next section, but for now let us suppose we do not have such a special q . Suppose we have built $q_1, \dots, q_m$, and we do not have $h_{m+1,m} = 0$. If $h_{m+1,m}$ is small, we might reasonably hope that we are close to an invariant subspace and the eigenvalues of H_m are close to eigenvalues of A . It turns out that even if $h_{m+1,m}$ is not small, some of the eigenvalues of H_m may be good approximations to eigenvalues of A .

Theorem 6.3.16 *Let Q_m , H_m , and $h_{m+1,m}$ be generated by the Arnoldi process, so that (6.3.13) holds. Let μ be an eigenvalue of H_m with associated eigenvector x , normalized so that $\|x\|_2 = 1$. Let $v = Q_m x \in \mathbb{C}^n$ (also with $\|v\|_2 = 1$). Then*

$$\|Av - \mu v\|_2 = |h_{m+1,m}| \|x_m\|, \quad (6.3.17)$$

where x_m denotes the m th (and last) component of x .

Exercise 6.3.18 Use (6.3.13) to prove Theorem 6.3.16. □

The vector v introduced in Theorem 6.3.16 is called a *Ritz vector* of A associated with the subspace $\mathcal{K}_m(A, q) = \text{span}\{q_1, \dots, q_m\}$, so called because it is a Rayleigh-Ritz-Galerkin approximation to an eigenvector of A (See Exercise 6.3.35). The scalar μ is called the *Ritz value* associated with v . The pair (μ, v) is called a *Ritz pair*.

If (μ, v) is an eigenpair of A , then the residual $Av - \mu v$ is zero. If (μ, v) is not an eigenpair, the residual norm $\|Av - \mu v\|_2$ will not be zero, but it will be close to zero if (μ, v) is a good approximation to an eigenpair. Conversely, if $\|Av - \mu v\|_2$ is small, we have reason to expect that (μ, v) is close to an eigenpair of A . It turns out that the relationship is not so simple: if the eigenpair that is being approximated is ill conditioned (See Section 6.5), a small residual does not guarantee a good approximation. Nevertheless, the residual norm does give some indication of the quality of the approximation: A small residual guarantees that (μ, v) is an exact eigenpair of a matrix that is close to A , as Exercise 6.3.36 shows. Finally, as we shall see in Section 6.5, all eigenvalues of normal matrices are well conditioned. Thus if A

is normal, a small residual guarantees that μ is near an eigenvalue of A . Recall that the class of normal matrices includes all real, symmetric matrices and other important classes of matrices.

Theorem 6.3.16 shows that the residual norm of the Ritz pair (μ, v) may be small even if $h_{m+1,m}$ is not, if the last entry of the eigenvector x is small. If m is not large, it is an easy matter to compute the eigenvalues and eigenvectors of H_m and check the quantities $|h_{m+1,m}| |x_m|$ for all eigenvectors x . If any of the residuals is small, we have reason to hope that we have a good approximation.

Experience has shown that some of the Ritz values will be good approximations to eigenvalues of A long before m is large enough that $h_{m+1,m}$ becomes small. Typically the eigenvalues on the periphery of the spectrum are approximated first, as the following example illustrates. See also Exercise 6.3.26.

Example 6.3.19 Using the following sequence of MATLAB instructions we built a random complex sparse matrix of dimension 144.

```
G = numgrid('N', 14);
B = delsq(G);
A = sprandn(B) + i*sprandn(B);
```

Since this example matrix is only of modest size, we can easily compute its eigenvalues by `eigvals = eig(full(A))`. We implemented Algorithm 6.3.6 in MATLAB and ran it for 40 steps, using the vector of all ones as the starting vector q , to get 40 orthonormal vectors and a 40×40 upper Hessenberg matrix H_{40} . We computed the eigenvalues and eigenvectors of H_{40} using `eig`. In Figure 6.1 we have plotted the true eigenvalues as pluses and the eigenvalues of H_{40} (Ritz values) as circles. The left-hand plot shows all of the Ritz values. We observe that many

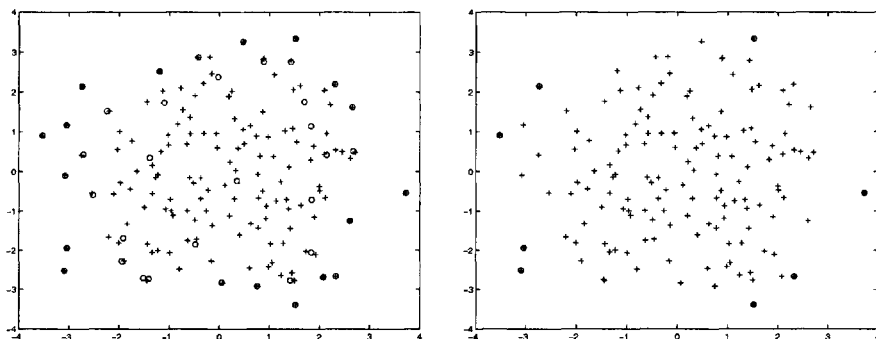


Fig. 6.1 Eigenvalues and Ritz values of a random sparse matrix

of the eigenvalues on the periphery of the spectrum are well approximated by Ritz values, while none of the interior eigenvalues are approximated well.

For each of the 40 Ritz pairs we computed the residual norm using the formula $|h_{41,40}| |x_{40}|$ from Theorem 6.3.16. The eight smallest residuals are listed in Table 6.1. In this example $h_{41,40} \approx 2.4$, so the smallness of each of the residuals is

residual ($ h_{41,40} x_{40} $)	true error
1.1×10^{-04}	1.3×10^{-06}
1.5×10^{-04}	6.0×10^{-06}
1.4×10^{-03}	2.5×10^{-04}
1.4×10^{-03}	1.2×10^{-04}
1.7×10^{-03}	7.1×10^{-05}
2.3×10^{-03}	1.8×10^{-04}
6.0×10^{-03}	5.9×10^{-04}
6.4×10^{-03}	1.0×10^{-03}

Table 6.1 Best eight Ritz pair residuals and corresponding Ritz value errors

attributable entirely to smallness of x_{40} , the last component of the eigenvector. Since we know the true eigenvalues in this case, we can compute the actual errors in the Ritz values. These are also shown in Table 6.1. In this benign case the errors are even smaller than the residuals. The eight best Ritz values are plotted, along with all 144 eigenvalues, in the right hand plot in Figure 6.1. $\square$

The reasons for the outer eigenvalues being approximated best are explored in Exercises 6.3.37 and 6.3.41.

Since the peripheral eigenvalues are found first, the Arnoldi process is not directly useful for computing interior eigenvalues. However, if the matrix A is not prohibitively large, the shift-and-invert strategy, which we introduced earlier in this section, can be combined with Arnoldi for this purpose. If we wish to find interior eigenvalues near the target τ , we can apply the Arnoldi process to $(A - \tau I)^{-1}$, whose peripheral eigenvalues $\mu_i = (\lambda_i - \tau)^{-1}$ correspond to the eigenvalues λ_i that are closest to τ . For each μ_i that we compute, we easily obtain $\lambda_i = \tau + 1/\mu_i$. The eigenvectors of $(A - \tau I)^{-1}$ are the same as the eigenvectors of A (Exercise 6.3.29).

The Symmetric Lanczos Process

When A is Hermitian, that is, $A = A^*$, the Arnoldi process takes a much simpler form. In most Hermitian applications the matrix is in fact real, so let's just assume that $A \in \mathbb{R}^{n \times n}$ and $A = A^T$. Then all of the eigenvalues are real, and so are the eigenvectors. More precisely, $\mathbb{R}^n$ has an orthonormal basis consisting of eigenvectors of A . Given a starting vector $q \in \mathbb{R}^{n \times n}$, after m steps of the Arnoldi process we have

$$AQ_m = Q_m T_m + q_{m+1} h_{m+1,m} e_m^T, \quad (6.3.20)$$

where Q_m is a real matrix with orthonormal columns, and T_m is a real upper Hessenberg matrix. The algorithm is simplified because T_m inherits the symmetry of A .

Exercise 6.3.21 Show that $T_m = Q_m^T A Q_m$, and T_m is symmetric if A is. Deduce that T_m is tridiagonal. $\square$

Since T_m is tridiagonal, most of the t_{jk} in the Arnoldi formula

$$\tilde{q}_{k+1} = Aq_k - \sum_{j=1}^k q_j t_{jk}$$

are zero; there are only two nonzero terms in the sum. If we introduce the notation

$$T_m = \begin{bmatrix} \alpha_1 & \beta_1 & & & \\ \beta_1 & \alpha_2 & & & \\ & & \ddots & & \\ & & & \ddots & \\ & & & & \beta_{m-1} & \alpha_m \end{bmatrix},$$

we obtain the three-term recurrence

$$\tilde{q}_{k+1} = Aq_k - q_k \alpha_k - q_{k-1} \beta_{k-1},$$

or, normalizing $\tilde{q}_{k+1}$,

$$q_{k+1} \beta_k = Aq_k - q_k \alpha_k - q_{k-1} \beta_{k-1}. \quad (6.3.22)$$

This special case of the Arnoldi process is called the *symmetric Lanczos process* [47].

Symmetric Lanczos Process without Reorthogonalization. Given $A = A^T \in \mathbb{R}^{n \times n}$ and a nonzero starting vector $q \in \mathbb{R}^n$, this algorithm produces orthonormal $q_1, q_2, \dots, q_m$ such that $\text{span}\{q_1, \dots, q_k\} = \text{span}\{q, Aq, \dots, A^{k-1}q\}$ for $k = 1, \dots, m$.

$$\begin{aligned} & q_1 = q / \|q\|_2 \\ & \text{for } k = 1, \dots, m-1 \\ & \quad \begin{cases} q_{k+1} \leftarrow Aq_k \\ \alpha_k \leftarrow q_k^T q_{k+1} \\ q_{k+1} \leftarrow q_{k+1} - q_k \alpha_k \\ \text{if } k > 1, \text{ then } q_{k+1} \leftarrow q_{k+1} - q_{k-1} \beta_{k-1} \\ \beta_k \leftarrow \|q_{k+1}\|_2 \\ \text{if } \beta_k = 0 \\ \quad \begin{cases} \text{set flag (span}\{q_1, \dots, q_k\} \text{ is invariant under } A) \\ \text{exit} \end{cases} \\ q_{k+1} \leftarrow q_{k+1} / \beta_k \end{cases} \end{aligned} \quad (6.3.23)$$

The coefficient α_k is computed as an inner product, just as in the Arnoldi process, but there is no need to compute β_{k-1} in this way, because it was already computed as the scaling factor β_k on the previous step. This is a consequence of symmetry of T_m .

More significant consequences of symmetry are that the algorithm is much cheaper and has much lower storage requirements, if it is executed without reorthogonalization, as shown in (6.3.23). If only eigenvalues are wanted, only the coefficients α_k and β_k need to be saved, as these are nonzero entries of T_m . At each step only the

two most recently computed vectors are needed; the others can be discarded. Thus the storage requirement is much less. At the k th step, only one inner product is computed instead of k , so the work is much less. For more information about this way of using the Lanczos process, see the books by Cullum and Willoughby [11].

The principle difficulty of using the symmetric Lanczos process without reorthogonalization is that the orthonormality of the q_k gradually deteriorates due to roundoff errors (Exercise 6.3.28.) If, as is usually the case, we are interested in maintaining orthonormality, then we must keep all the q_k and orthogonalize each new vector against them (twice). Thus the storage and computation savings are nullified. Notice that we also need to save all the q_k if we want to compute *eigenvectors* of A .

Additional Exercises

Exercise 6.3.24 Use MATLAB's test matrix `west0479` to demonstrate that similarity transformations can cause a great deal of fill in.

- (a) Show that the reduction to upper Hessenberg form fills in the entire upper part of the matrix. For example,

```
load west0479
A = west0479;
nnza = nnz(A)
spy(A)
H = hess(full(A)); % hess operates on full matrices.
nnzh = nnz(H);
spy(H)
```

- (b) Apply one iteration of the QR algorithm (with or without shift) to A (not H). You can do this quickly by using MATLAB's `qr` command to perform the QR decomposition. Make a spy plot of the result.

□

Exercise 6.3.25 The sparse data structure used by MATLAB lists the nonzero entries of A in a list s of length nz . In addition there are two lists of integers row and col , also of length nz , that specify where each entry belongs in A . For example, if the seventh entry in the list is $s(7) = 1.234 \times 10^5$, and $row(7) = 45$ and $col(7) = 89$, then $a_{45,89} = 1.234 \times 10^5$. (A more detailed description was given in Section 1.6.) Suppose A is stored in this way. Write a pseudocode algorithm that computes $y = Ax$, assuming the vector x (and also y) is stored in the conventional way. Your algorithm should make just one pass through the list s and perform $2nz$ flops in all.

□

Exercise 6.3.26 Write a MATLAB script that implements the Arnoldi algorithm with reorthogonalization. You may use Algorithm 6.3.6 as a pattern if you wish. Use a random vector (`q = randn(n,1)`) or the vector of all ones (`q = ones(n,1)`) as a starting vector. Use `eig` to compute the eigenvalues of the Hessenberg ma-

trix H_m that is generated. These are the Ritz values, some of which may be good estimates of eigenvalues. To test your code, generate a sparse matrix by

```
num = 14; % or other appropriate number
G = numgrid('N',num);
B = delsq(G);
A = sprandn(B) + i*sprandn(B);
```

as in Example 6.3.19. (Use `help numgrid`, `help delsq`, and so on for information about these functions.) Since this method uses a random number generator, you will get a different matrix from the one that was used in Example 6.3.19. You can get a larger or smaller test matrix by making `num` larger or smaller. (You will get a matrix of dimension $(\text{num} - 2)^2$.) Check the nonzero pattern of A by using `spy`. Using a random starting vector or other starting vector of your choice, run the Arnoldi process for 15, 30, 45, and 60 steps to get Hessenberg matrices of sizes 15, 30, 45, and 60. (Actually you can just do one run, because the smaller Hessenberg matrices are submatrices that can be extracted from H_{60} .) Calculate the Ritz values for each of the four cases (for example, `ritz = eig(h(1:15,1:15))`), and in each case plot the Ritz values and the true eigenvalues together. For example, you could use commands like these:

```
hold off
eigval = eig(full(A));
plot(real(eigval),imag(eigval),'r+')
hold on
plot(real(ritz),imag(ritz),'bo')
```

Notice that the quality of approximation improves as the number of steps is increased, and the outer eigenvalues are approximated best. $\square$

Exercise 6.3.27 Repeat Exercise 6.3.26 taking A to be the test matrix `west0479`.

```
load west0479
A = west0479;
```

$\square$

Exercise 6.3.28 This exercise explores the loss of orthogonality in the Arnoldi process without reorthogonalization. If you have not already done so, write a MATLAB script that implements the Arnoldi algorithm with reorthogonalization. Modify it so that the reorthogonalization can be turned on or off, depending on the value of some switching variable. For each of the following matrices, take 100 Arnoldi steps without reorthogonalization. Then compute the inner products $q_1^* q_k$ for $k = 2, \dots, 100$. Notice how the orthogonality deteriorates as k increases. Also compute $\|I_m - Q_m^* Q_m\|$ (with $m = 100$), which should be zero in theory. Now repeat the Arnoldi process with reorthogonalization and compute $\|I_m - Q_m^* Q_m\|$.

(a) Use the matrix `west0479`.

```
load west0479
A = west0479;
spy(A)
n = size(A,1)
```

- (b) Now try a discrete negative Laplacian matrix of dimension 324 (from the interior of a 20×20 square grid). This sparse matrix is symmetric.

```
A = delsq(numgrid('N',20));
n = size(A,1)
```

- (c) Now, to show that loss of orthogonality does not always occur, try a matrix like the one from Exercise 6.3.26.

```
B = delsq(numgrid('N',20));
A = sprandn(B) + i*sprandn(B);
n = size(A,1)
```

□

Exercise 6.3.29 Modify your code from Exercise 6.3.26 to do the shift-and-invert Arnoldi process. This requires only a few modifications. An LU decomposition of $(A - \tau I)$, where τ is the target shift, needs to be done once at the beginning:

```
[L,U] = lu(A-tau*speye(size(A)));
```

speye is a sparse identity matrix. Then whenever an operation of the form

$$q \leftarrow (A - \tau I)^{-1} q$$

needs to be done, it can be effected by the operations $q = L \backslash q$; $q = U \backslash q$. The eigenvalues that are computed are eigenvalues of $(A - \tau I)^{-1}$, so your code will need to transform them to eigenvalues of A . From $\mu = (\lambda - \tau)^{-1}$, we obtain $\lambda = \tau + 1/\mu$. Test your program by using it to calculate the ten eigenvalues of `west0479` that are closest to $\tau = 6 + 0i$. Take as many Arnoldi steps as you need to get the residuals down around 10^{-16} . Compute the residuals by the inexpensive method of Theorem 6.3.16. Compare your computed eigenvalues with the “true” eigenvalues computed by the QR algorithm (`lam = eig(full(west0479))`). Find the eigenvalues that are closest to 6 by using MATLAB’s `sort` command:

```
[dummy,order] = sort(lam-tau);
lam = lam(order);
```

□

Exercise 6.3.30 Prove Theorem 6.3.9 by induction on k . You might like to proceed as follows. Assuming that $q, Aq, A^2q, \dots, A^{m-1}q$ are linearly independent, we must show that $\text{span}\{q_1, \dots, q_k\} = \mathcal{K}_k(A, q)$ for $k = 1, \dots, m$.

- (a) Establish that the case $k = 1$ holds, i.e. $\text{span}\{q_1\} = \mathcal{K}_1(A, q)$.
 (b) To establish the general result, let $j < m$. You must show that if

$$\text{span}\{q_1, \dots, q_k\} = \mathcal{K}_k(A, q) \quad \text{for } k = 1, \dots, j,$$

then $\text{span}\{q_1, \dots, q_{j+1}\} = \mathcal{K}_{j+1}(A, q)$. As part of the induction hypothesis we have $\text{span}\{q_1, \dots, q_j\} = \mathcal{K}_j(A, q)$, so $q_j \in \mathcal{K}_j(A, q)$. Thus there exist constants $c_0, \dots, c_{j-1}$ such that

$$q_j = c_0 q + c_1 Aq + c_2 A^2 q + \dots + c_{j-1} A^{j-1} q. \quad (6.3.31)$$

Show that $c_{j-1} \neq 0$. (You can use $\text{span}\{q_1, \dots, q_{j-1}\} = \mathcal{K}_{j-1}(A, q)$ here.) Now multiply (6.3.31) by A to obtain an expression for Aq_j , which you can use in (6.3.3) to show that $\tilde{q}_{j+1} \neq 0$. Thus $h_{j+1,j} > 0$, and we can do the division in (6.3.5) to obtain q_{j+1} . Show that there exist constants $d_0, \dots, d_j$ such that

$$q_{j+1} = d_0 q + d_1 Aq + d_2 A^2 q + \dots + d_j A^j q. \quad (6.3.32)$$

Thus $q_{j+1} \in \mathcal{K}_{j+1}(A, q)$. Show that $\text{span}\{q_1, \dots, q_{j+1}\} \subseteq \mathcal{K}_{j+1}(A, q)$.

- (c) Continuing from part (b), show that $d_j \neq 0$ in (6.3.32). Using (6.3.32) and the induction hypothesis $\text{span}\{q_1, \dots, q_j\} = \mathcal{K}_j(A, q)$, show that $A^j q \in \text{span}\{q_1, \dots, q_{j+1}\}$ and hence $\mathcal{K}_{j+1}(A, q) \subseteq \text{span}\{q_1, \dots, q_{j+1}\}$. Conclude that $\text{span}\{q_1, \dots, q_{j+1}\} = \mathcal{K}_{j+1}(A, q)$. This completes the induction proof of part (a) of Theorem 6.3.9. Along the way we have proved part (b) as well.
- (d) Using relationships developed in the previous parts of this exercise, prove that if $q, Aq, A^2 q, \dots, A^m q$ are linearly dependent, then $\tilde{q}_{m+1} = 0$ and $h_{m+1,m} = 0$. Conversely, if $q, Aq, A^2 q, \dots, A^m q$ are linearly independent, then $\tilde{q}_{m+1} \neq 0$ and $h_{m+1,m} \neq 0$. This was already proved in part (b). Now complete the proof of Theorem 6.3.9, part (c).

□

Exercise 6.3.33 Prove Proposition 6.3.15 by two methods.

- (a) Show that if (6.3.13) is satisfied, then so is (6.3.11), which is equivalent to (6.3.3) combined with (6.3.5). Show that the coefficients h_{jk} in (6.3.3) are uniquely determined by the requirement that $\tilde{q}_{k+1}$ be orthogonal to $q_1, \dots, q_k$. Show that $h_{k+1,k}$ is uniquely determined by the requirements that it be positive and that $\|q_{k+1}\|_2 = 1$. Prove by induction that all q_j are uniquely determined by q_1 .
- (b) The second approach uses Krylov matrices, which are closely related to Krylov subspaces. Given $A \in \mathbb{C}^{n \times n}$, $x \in \mathbb{C}^n$, and $j \leq n$, the *Krylov matrix* $K_j(A, x) \in \mathbb{C}^{n \times j}$ is the matrix with columns $x, Ax, A^2 x, \dots, A^{j-1} x$.
- Let H_m be as in (6.3.13); that is, H_m is upper Hessenberg with positive subdiagonal entries. Show that the Krylov matrix $K_m(H_m, e_1)$ is upper triangular and has all of its main-diagonal entries positive.
 - Show that if (6.3.13) is satisfied, then $AQ_m y = Q_m H_m y$ for all $y \in \mathbb{C}^m$ satisfying $e_m^T y = 0$.
 - Still assuming (6.3.13) and using the result from part (ii), prove by induction on j that $A^j Q_m e_1 = Q_m H_m^j e_1$ for $j = 0, 1, \dots, m-1$. Deduce that $K_m(A, q_1) = Q_m K_m(H_m, e_1)$.
 - Using the uniqueness of QR decompositions (Theorem 3.4.8), prove that Q_m is uniquely determined by q_1 .

The method of part (b) should be compared with that of Exercise 5.7.46, in which the Implicit- Q Theorem was proved. Proposition 6.3.15 reduces to the Implicit- Q Theorem in the case $m = n$. $\square$

Exercise 6.3.34 Count the flops required to transform a matrix to upper Hessenberg form by running the Arnoldi process to completion (n steps of Algorithm 6.3.6). Count the cost of the orthogonalization separately from the matrix-vector multiplications Aq_k . If the vectors q_k are to be orthonormal to working precision, we must reorthogonalize, as shown in Algorithm 6.3.6. Show that if we do this, the cost of the orthogonalization alone is more than the cost of the reduction to upper Hessenberg form using reflectors (Algorithm 5.5.2). Assuming A is a dense matrix, show that the total cost of the Arnoldi algorithm is about $6n^3$ flops. $\square$

Exercise 6.3.35 Let $A \in \mathbb{C}^{n \times n}$, and let $\mathcal{S}$ be a k -dimensional subspace of $\mathbb{C}^n$. Then a vector $v \in \mathcal{S}$ is called a *Ritz vector* of A from $\mathcal{S}$ if and only if there is a $\mu \in \mathbb{C}$ such that the *Rayleigh-Ritz-Galerkin* condition

$$Av - \mu v \perp \mathcal{S}$$

holds, that is, $\langle Av - \mu v, s \rangle = 0$ for all $s \in \mathcal{S}$. The scalar μ is called the *Ritz value* of A associated with v . Let $q_1, \dots, q_k$ be an orthonormal basis of $\mathcal{S}$, let $Q = [q_1 \cdots q_k]$, and let $B = Q^*AQ \in \mathbb{C}^{k \times k}$.

- Since $v \in \mathcal{S}$, there is a unique $x \in \mathbb{C}^k$ such that $v = Qx$. Show that v is a Ritz vector of A with associated Ritz value μ if and only if x is an eigenvector of B with associated eigenvalue μ . In particular, there are k Ritz values of A associated with $\mathcal{S}$, namely the k eigenvalues of B .
- Show that if v is a Ritz vector with Ritz value μ , then $\mu = v^*Av/v^*v$. That is, μ is the Rayleigh quotient of v . (More generally, we can think of $B = Q^*AQ$ as a $k \times k$ matrix *Rayleigh quotient*.)
- Suppose Q_m and H_m are generated by applying the Arnoldi process to A . Show that the eigenvalues of H_m are Ritz values of A associated with the subspace spanned by the columns of Q_m .

$\square$

Exercise 6.3.36 Let (μ, v) be an approximate eigenpair of A with $\|v\|_2 = 1$, and let $r = Av - \mu v$ (the residual). Let $\epsilon = \|r\|_2$ and $E = -rv^*$. Show that (μ, v) is an eigenpair of $A + E$, and $\|E\|_2 = \epsilon$. This shows that if the residual norm $\|r\|_2$ is small, then (μ, v) is an exact eigenpair of a matrix that is close to A . Thus (μ, v) is a good approximate eigenpair of A in the sense of backward error. $\square$

Exercise 6.3.37

- Given $A \in \mathbb{C}^{n \times n}$ and a polynomial $p(x) = a_0 + a_1x + a_2x^2 + \cdots + a_jx^j$, we define the matrix $p(A) \in \mathbb{C}^{n \times n}$ by $p(A) = a_0I + a_1A + a_2A^2 + \cdots + a_jA^j$.

Show that a vector v is in the Krylov subspace $\mathcal{K}_m(A, q)$ if and only if v can be expressed as $p(A)q$, where p is a polynomial of degree $m - 1$ or less. Thus $\mathcal{K}_m(A, q) = \{p(A)q \mid p \in \mathcal{P}_{m-1}\}$, where $\mathcal{P}_{m-1}$ denotes the set of polynomials of degree less than or equal to $m - 1$.

- (b) Show that if v is an eigenvector of A with eigenvalue λ , then v is an eigenvector of $p(A)$ with eigenvalue $p(\lambda)$.
- (c) Suppose A is a semisimple matrix with linearly independent eigenvectors $v_1, \dots, v_n$ and associated eigenvalues $\lambda_1, \dots, \lambda_n$. Then $q = c_1 v_1 + c_2 v_2 + \dots + c_n v_n$, for some scalars $c_1, \dots, c_n$, which are not known in advance. If q was chosen “at random,” then (typically) none of the c_i will be extremely small, and q will have a significant contribution from each of the eigenvectors. Let us assume that this is the case, and that $\|v_j\| = 1$ for all j . Show that if $v = p(A)q$, then

$$v = c_1 p(\lambda_1) v_1 + c_2 p(\lambda_2) v_2 + \dots + c_n p(\lambda_n) v_n.$$

Show that if there is a $p \in \mathcal{P}_{m-1}$ such that, for some j , $|p(\lambda_j)|$ is much larger $|p(\lambda_i)|$ for all $i \neq j$, then the Krylov subspace $\mathcal{K}_m(A, q)$ will contain a vector that is close to the eigenvector v_j . Under these circumstances we normally would expect that one of the Ritz pairs from $\mathcal{K}_m(A, q)$ will approximate the eigenpair (λ_j, v_j) well.

- (d) Suppose λ_j is well separated from the other eigenvalues of A . How might one build a polynomial $p(z) = (z - z_1)(z - z_2) \cdots (z - z_{m-1})$ such that $|p(\lambda_j)| \gg |p(\lambda_i)|$ for all $i \neq j$? (Think about the placement of the zeros z_i .) How would the task be more difficult if λ_j were not well separated from the other eigenvalues of A ? For these reasons the Arnoldi process tends to find eigenvalues on the periphery of the spectrum first, since these are better isolated from the rest of the eigenvalues. These questions are studied more closely for symmetric matrices in Exercise 6.3.41.

□

Exercise 6.3.38 In this Exercise you will develop the basic properties of Chebyshev polynomials, which are perhaps the most useful polynomials in approximation theory. In Exercise 6.3.41 you will use the Chebyshev polynomials to study the rate of convergence of the symmetric Lanczos process. Later on (Exercise 7.8.13), you will use them in the analysis of the conjugate-gradient algorithm.

For $x \in [-1, 1]$, define an auxiliary variable $\theta \in [0, \pi]$ by $x = \cos \theta$. The value of θ is uniquely determined by x ; in fact $\theta = \arccos x$. Now for $m = 0, 1, 2, \dots$ define a function T_m on $[-1, 1]$ by

$$T_m(x) = \cos m\theta.$$

- (a) Show that $|T_m(x)| \leq 1$ for all $x \in [-1, 1]$, $T_m(1) = 1$, $T_m(-1) = (-1)^m$, $|T_m(x)| = 1$ at $m + 1$ distinct points in $[-1, 1]$, and $T_m(x) = 0$ at m distinct points in $(-1, 1)$.

(b) Use the trigonometric identity

$$\cos(\alpha + \beta) + \cos(\alpha - \beta) = 2 \cos \alpha \cos \beta$$

to show that

$$T_{m+1}(x) + T_{m-1}(x) = 2xT_m(x) \quad m = 1, 2, 3, \dots$$

This yields the recursion

$$T_{m+1}(x) = 2xT_m(x) - T_{m-1}(x) \quad m = 1, 2, 3, \dots \quad (6.3.39)$$

- (c) Determine $T_0(x)$ and $T_1(x)$ directly from the definition, then use (6.3.39) to calculate $T_2(x)$, $T_3(x)$, and $T_4(x)$. Note that each is a polynomial in x and can therefore be extended in a natural way beyond the interval $[-1, 1]$. Graph the polynomials T_2 , T_3 , and T_4 , focusing on the interval $[-1, 1]$, but notice that they grow rapidly once x leaves the interval.
- (d) Using the recursion (6.3.39), prove by induction on m that for all m , T_m is a polynomial of degree m . (From part (a) we know that T_m has all m of its zeros in $(-1, 1)$. Therefore T_m must grow rapidly once x leaves that interval.)
- (e) Not only does T_m grow rapidly once x leaves $[-1, 1]$. For a given x not in $[-1, 1]$, $T_m(x)$ grows rapidly as a function of m . To show this, we need another characterization of the Chebyshev polynomials. For $x \in [1, \infty)$ define an auxiliary variable $t \in [0, \infty)$ by $x = \cosh t$. (Recall that $\cosh t$, the *hyperbolic cosine* of t , is defined by $\cosh t = \frac{1}{2}(e^t + e^{-t})$.) For $m = 0, 1, 2, \dots$ define a function $\tilde{T}_m$ on $[1, \infty)$ by

$$\tilde{T}_m(x) = \cosh mt.$$

Use the hyperbolic identity

$$\cosh(\alpha + \beta) + \cosh(\alpha - \beta) = 2 \cosh \alpha \cosh \beta$$

to show that

$$\tilde{T}_{m+1}(x) + \tilde{T}_{m-1}(x) = 2x\tilde{T}_m(x) \quad m = 1, 2, 3, \dots,$$

hence

$$\tilde{T}_{m+1}(x) = 2x\tilde{T}_m(x) - \tilde{T}_{m-1}(x) \quad m = 1, 2, 3, \dots,$$

which is exactly the recursion (6.3.39). Compute $\tilde{T}_0$ and $\tilde{T}_1$ directly from the definition, and prove by induction on m that $\tilde{T}_m = T_m$ for all m . Thus

$$T_m(x) = \cosh mt, \quad \text{where } x = \cosh t. \quad (6.3.40)$$

- (f) If $x > 1$, then $x = \cosh t$, where $t > 0$. Let $\rho = e^t > 1$. Use (6.3.40) to show that $T_m(x) \geq \frac{1}{2}\rho^m$. Thus $T_m(x)$ grows exponentially as a function of m .

- (g) Use MATLAB (which has built-in functions $\cosh$ and acosh) to compute $t = \cosh^{-1} x$ and $\rho = e^t$ when $x = 1.1$. compute $T_{10}(1.1)$, $T_{20}(1.1)$, and $T_{30}(1.1)$. Compare the ratio $T_{30}(1.1)/T_{20}(1.1)$ with e^{10t} . To what can you attribute the good agreement?
- (h) Show that $T_m(-x) = (-1)^m T_m(x)$ for all m and x . Therefore we have the same growth of $|T_m(x)|$ for $x \in (-\infty, -1)$ as for $x \in (1, \infty)$.

□

Exercise 6.3.41 This exercise makes use of the Chebyshev polynomials, introduced in the previous exercise, to pursue the theme of Exercise 6.3.37 in greater detail. Let $A \in \mathbb{R}^{n \times n}$ be a symmetric matrix with eigenvalues $\lambda_1 > \lambda_2 \geq \cdots \geq \lambda_n$, with $\lambda_2 > \lambda_n$. We can obtain polynomials p_m with Chebyshev behavior on the interval $[\lambda_n, \lambda_2]$ by performing a transformation that maps $[\lambda_n, \lambda_2]$ onto $[-1, 1]$. These polynomials will have the property that $p_m(\lambda_1)$ is large and $p_m(\lambda_j)$ is bounded by 1 for $j \neq 1$.

- (a) Show that the transformation

$$w = 1 + 2 \frac{x - \lambda_2}{\lambda_2 - \lambda_n}$$

maps $[\lambda_n, \lambda_2]$ onto $[-1, 1]$.

- (b) Define a polynomial $p_m \in \mathcal{P}_m$ by

$$p_m(x) = T_m(w) = T_m\left(1 + 2 \frac{x - \lambda_2}{\lambda_2 - \lambda_n}\right).$$

Show that $|p_m(\lambda_i)| \leq 1$ for $i \geq 2$ and

$$|p_m(\lambda_1)| \geq \frac{1}{2} \rho^m,$$

where

$$\rho \geq 1 + 2 \frac{\lambda_1 - \lambda_2}{\lambda_2 - \lambda_n} > 1. \quad (6.3.42)$$

Use part (f) of Exercise 6.3.38.

- (c) Let $v_1, \dots, v_n$ denote orthonormal eigenvectors of A associated with the eigenvalues $\lambda_1, \dots, \lambda_n$, respectively. Let q be a starting vector for the symmetric Lanczos algorithm with $\|q\|_2 = 1$, and suppose $q = c_1 v_1 + c_2 v_2 + \cdots + c_n v_n$ with $c_1 \neq 0$. Define $\tilde{p}_m$ by $\tilde{p}_m(x) = p_m(x)/(c_1 p_m(\lambda_1))$, where p_m is as defined in part (b). Let $w_m = \tilde{p}_{m-1}(A)q \in \mathcal{K}_m(A, q)$. Show that there is a constant C such that

$$\|w_m - v_1\|_2 \leq C\rho^{-m}, \quad m = 1, 2, 3, \dots,$$

where ρ is as in (6.3.42). Thus, for large m , the Krylov subspace $\mathcal{K}_m(A, q)$ contains a vector that is close to the eigenvector v_1 .

- (d) Adapt the arguments given above to show that for large m , $\mathcal{K}_m(A, q)$ contains vectors that are close to v_n , the eigenvector associated with λ_n . Assume $\lambda_1 > \lambda_{n-1} > \lambda_n$ and $c_n \neq 0$.
- (e) Suppose $\lambda_1 > \lambda_2 > \lambda_3 > \lambda_n$. For $m = 1, 2, 3, \dots$, define a polynomial $\hat{p}_m \in \mathcal{P}_m$ by

$$\hat{p}_m(x) = (x - \lambda_1)T_{m-1} \left(1 + 2 \frac{x - \lambda_3}{\lambda_3 - \lambda_n} \right).$$

Use $\hat{p}_m$ to show that for sufficiently large m , the Krylov subspace $\mathcal{K}_m(A, q)$ contains vectors that are close to v_2 . Assume $c_2 \neq 0$.

- (f) Suppose $\lambda_2 > \lambda_3 > \lambda_4 > \lambda_n$. Construct polynomials that can be used to show that for sufficiently large m , $\mathcal{K}_m(A, q)$ contains vectors that are close to v_3 , assuming $c_3 \neq 0$.

□

Exercise 6.3.43 This exercise is for those who are familiar with the three-term recurrence (Stieltjes process) for generating orthogonal polynomials. First note that the Lanczos procedure is governed by a three-term recurrence (6.3.22). Then show that the three-term recurrence for generating orthogonal polynomials can be viewed as the Lanczos algorithm applied to a certain linear operator acting on an infinite-dimensional space. (Note: The recurrence (6.3.39) for the Chebyshev polynomials is a special case.) □

6.4 EIGENVALUES OF LARGE, SPARSE MATRICES, II

This section, which builds on the previous one, introduces some of the most effective known methods for computing eigenvalues of large matrices, including the implicitly restarted Arnoldi process and the Jacobi-Davidson method. For more information see [3].

The Implicitly Restarted Arnoldi Process

The rate at which the Arnoldi process begins to give good approximations to eigenpairs depends very much upon the choice of starting vector q . In most cases we cannot hope to find a good q at the outset; commonly q is chosen at random, due to a lack of information. However, once we have taken a few Arnoldi steps, we have new information that we might be able to use to find a new vector $\hat{q}$ that would have been a better starting vector. Supposing we can find such a $\hat{q}$, might it be worthwhile to start a new Arnoldi process with $\hat{q}$ instead of continuing the old one? If so, might it be worthwhile to restart repeatedly, with a better starting vector each time?

Indeed, repeated restarts are worthwhile, and that is exactly what the implicitly restarted Arnoldi process (IRA) does. After a short Arnoldi run, it restarts with a new vector, does another short Arnoldi run, restarts again, and so on. Because of the nature of the process by which new starting vectors are chosen, it turns out not to be necessary to start each new run from scratch, rather it can be picked up in midstream, so to speak. For this reason the restarts are called *implicit*.

After a number of implicit restarts, the process terminates with a low-dimensional invariant subspace containing the desired eigenvectors. It is then a simple matter to extract those eigenvectors and the associated eigenvalues.

MATLAB has a function called `eigs`, which computes a few eigenvalues of a sparse matrix by IRA. See Exercise 6.4.19. A Fortran implementation of IRA called ARPACK [49] is available for free.

A major advantage of restarts is that less storage space is needed. If we restrict ourselves to short Arnoldi runs of length m , say, then we only have to store about m vectors. Each time we restart, we free up memory. Furthermore, since only a few vectors are kept, it is not too expensive to reorthogonalize them as necessary to ensure that they remain truly orthonormal to working precision.

Now that we have the basic idea of the process and have noted some potential advantages, an obvious question arises: What constitutes a good starting vector? To begin with an extreme example, suppose we are interested in a certain eigenvector. Then it would be nice to start the Arnoldi process with that eigenvector. If we are so lucky as to start with such a q , then the process will terminate in one step with the eigenvector q_1 and an associated eigenvalue. In many cases we are interested in more than one eigenpair. Suppose we are able to start with a q that is a linear combination of a small number of eigenvectors, say $q = c_1 v_1 + \cdots + c_k v_k$ with $k \ll n$. Then the Arnoldi process will terminate in k or fewer steps with (typically) $\text{span}\{q_1, \dots, q_k\}$ equal to the invariant subspace $\text{span}\{v_1, \dots, v_k\}$ (see Exercise 6.4.20).

This suggests that we try to build a q that is a linear combination of just a few desired eigenvectors. To keep the discussion simple, let us assume that A is a semisimple matrix with linearly independent eigenvectors $v_1, \dots, v_n$ and associated eigenvalues $\lambda_1, \dots, \lambda_n$. For definiteness, let us suppose we are looking for the k eigenvalues of largest modulus and their associated eigenvectors. Let us number the eigenvalues so that $|\lambda_1| \geq |\lambda_2| \geq \cdots \geq |\lambda_n|$ and assume that $|\lambda_k| > |\lambda_{k+1}|$. Then the desired eigenvectors are $v_1, \dots, v_k$.

Since $v_1, \dots, v_n$ form a basis for $\mathbb{C}^n$, we can certainly write, for any q ,

$$q = c_1 v_1 + c_2 v_2 + \cdots + c_n v_n$$

for some uniquely determined but unknown $c_1, c_2, \dots, c_n$. If we choose q at random, it is likely to have significant components in the directions of all eigenvectors. That is, none of the c_j will be zero or exceptionally close to zero. This is far from what we want. Since the desired eigenvectors are $v_1, \dots, v_k$, we would like to have $c_{k+1} = c_{k+2} = \cdots = c_n = 0$. Our task, then, is to find a new vector

$$\hat{q} = \hat{c}_1 v_1 + \hat{c}_2 v_2 + \cdots + \hat{c}_n v_n,$$

in which $\hat{c}_1, \dots, \hat{c}_k$ have been augmented and $\hat{c}_{k+1}, \dots, \hat{c}_n$ have been diminished. Suppose we take $\hat{q} = p(A)q$, where p is some polynomial. Then (see Exercise 6.3.37)

$$\hat{q} = c_1 p(\lambda_1) v_1 + c_2 p(\lambda_2) v_2 + \dots + c_n p(\lambda_n) v_n. \quad (6.4.1)$$

If we can choose p so that $p(\lambda_1), \dots, p(\lambda_k)$ are large in comparison with $p(\lambda_{k+1}), \dots, p(\lambda_n)$, we will have made progress. This is what the implicitly restarted Arnoldi process seeks to do.

We now describe an iteration of the implicitly restarted Arnoldi process (IRA). If we want to find k eigenvalues, we make Arnoldi runs of length $m = k + j$, where j is comparable to k . A popular choice is $j = k$. Starting from the vector q , after m steps we have generated $Q_m = [q_1 \ \dots \ q_m]$ with orthonormal columns and upper Hessenberg H_m such that

$$AQ_m = Q_m H_m + q_{m+1} h_{m+1,m} e_m^T, \quad (6.4.2)$$

as in (6.3.13). Since m is small, we easily compute (by the QR algorithm) the m eigenvalues of H_m . These are the Ritz values of A associated with the subspace $\mathcal{R}(Q_m) = \text{span}\{q_1, \dots, q_m\}$. Let us call them $\mu_1, \dots, \mu_m$ and order them so that $|\mu_1| \geq |\mu_2| \geq \dots \geq |\mu_m|$. The largest ones, $\mu_1, \dots, \mu_k$, are estimates of the k largest eigenvalues of A (the desired eigenvalues), while $\mu_{k+1}, \dots, \mu_m$ approximate other (undesired) parts of the spectrum. Although these may be poor approximations, they at least give us a crude picture of the location of the spectrum of A .

IRA then performs j iterations of the QR algorithm on H_m , using j shifts $\nu_1, \dots, \nu_j$ in the region of spectrum that we want to suppress. The most popular choice is to take $\nu_1 = \mu_{k+1}, \nu_2 = \mu_{k+1}, \dots, \nu_j = \mu_m$. If we make this choice, we are doing the *exact shift* variant of IRA. The QR iterations are inexpensive, because m is small. Their combined effect is a unitary similarity transformation

$$\hat{H}_m = V_m^{-1} H_m V_m, \quad (6.4.3)$$

where

$$p(H_m) = V_m R_m, \quad (6.4.4)$$

V_m is unitary, R_m is upper triangular, and p is a polynomial of degree j with zeros $\nu_1, \dots, \nu_j$:

$$p(z) = (z - \nu_1)(z - \nu_2) \dots (z - \nu_j),$$

as was shown in Exercise 6.2.36. The QR iterations preserve upper Hessenberg form, so $\hat{H}_m$ is also upper Hessenberg. Let $\hat{Q}_m = Q_m V_m$, and in particular let $\hat{q}_1$ be the first column of $\hat{Q}_m$.

The next iteration of IRA consists of another Arnoldi run of m steps, starting from $\hat{q}_1$. However, as we have already indicated, we need not start the Arnoldi run from scratch. To see why this is so, multiply (6.4.2) by V_m on the right, and use (6.4.3) to obtain

$$A\hat{Q}_m = \hat{Q}_m \hat{H}_m + q_{m+1} h_{m+1,m} e_m^T V_m. \quad (6.4.5)$$

Exercise 6.4.21 shows that the row vector $e_m^T V_m$ has exactly $m - j - 1$ leading zeros. Thus if we drop the last j entries from this vector, we obtain a vector of the form

βe_k^T , where β is some nonzero scalar. Therefore, if we drop the last j columns from equation (6.4.5), we obtain $A\hat{Q}_k = \hat{Q}_{k+1}\hat{H}_{k+1,k} + q_{m+1}h_{m+1,m}\beta e_k^T$ or

$$A\hat{Q}_k = \hat{Q}_k\hat{H}_k + \tilde{q}_{k+1}\tilde{h}_{k+1,k}e_k^T + q_{m+1}h_{m+1,m}\beta e_k^T, \quad (6.4.6)$$

where we have used $\tilde{q}_{k+1}$ to denote the $(k+1)$ st column of $\hat{Q}_m$ and $\tilde{h}_{k+1,k}$ to denote the $(k+1, k)$ entry of $\hat{H}_{k+1,k}$. Define $\hat{q}_{k+1} = \gamma(\tilde{q}_{k+1}\tilde{h}_{k+1,k} + q_{m+1}h_{m+1,m}\beta)$, where γ is a positive scalar chosen so that $\|\hat{q}_{k+1}\|_2 = 1$.

Exercise 6.4.7 Show that q_{m+1} is orthogonal to $\tilde{q}_{k+1}$. Use this fact to show that $\tilde{q}_{k+1}\tilde{h}_{k+1,k} + q_{m+1}h_{m+1,m}\beta$ is nonzero. Thus there exists a $\gamma > 0$ such that $\|\hat{q}_{k+1}\|_2 = 1$. Show that $\hat{q}_{k+1}$ is orthogonal to $\hat{q}_1, \dots, \hat{q}_k$. $\square$

Let $\hat{h}_{k+1,k} = 1/\gamma$. Then (6.4.6) becomes

$$A\hat{Q}_k = \hat{Q}_k\hat{H}_k + \hat{q}_{k+1}\hat{h}_{k+1,k}e_k^T, \quad (6.4.8)$$

which is identical to (6.3.13), except for the hats on the symbols and the m having been replaced by k . Thus Proposition 6.3.15 implies that the columns of $\hat{Q}_k$ are exactly the vectors that would be built by the Arnoldi process, starting from $\hat{q}_1$.

Thus IRA need not start from scratch when building up an Arnoldi sequence from $\hat{q}_1$. Instead it can extract $\hat{Q}_k$ and $\hat{H}_k$ (submatrices of $\hat{Q}_m = Q_m V_m$ and $\hat{H}_m = V_m^{-1} H_m V_m$, respectively) and start with step k . Thus $k-1$ Arnoldi steps are avoided.

IRA is summarized in the following algorithm.

Implicitly Restarted Arnoldi (IRA) Process. Given $A \in \mathbb{C}^{n \times n}$ and a nonzero starting vector $q \in \mathbb{R}^n$, this algorithm produces orthonormal $q_1, q_2, \dots, q_k$ that span an invariant subspace. If an invariant subspace is not found in *imax* iterations, the algorithm terminates with an error flag.

$$\begin{aligned} & q_1 = q/\|q\|_2 \\ & \text{take } k \text{ Arnoldi steps (Algorithm 6.3.6) to produce } Q_k \text{ and } H_k \\ & \text{for } ii = 1, 2, \dots, \text{imax} \\ & \quad \left[\begin{array}{l} \text{take another } j \text{ Arnoldi steps (Algorithm 6.3.6)} \\ \quad \text{to produce } Q_m \text{ and } H_m, m = k + j. \\ \text{if } |h_{k+1,k}| < \text{tol, exit } (\mathcal{R}(Q_k) \text{ is invariant under } A) \\ \text{find } j \text{ shifts } \nu_1, \dots, \nu_j \\ \text{take } j \text{ steps of the } QR \text{ algorithm with shifts } \nu_1, \dots, \nu_j \\ \quad \text{to obtain transformation matrix } V_m \\ H_m \leftarrow V_m^{-1} H_m V_m \\ Q_m \leftarrow Q_m V_m \\ Q_k \leftarrow Q_m(:, 1:k) \\ H_k \leftarrow H_m(1:k, 1:k) \end{array} \right. \quad (6.4.9) \\ & \text{set flag (failed to converge in } \text{imax} \text{ iterations)} \end{aligned}$$

In the interest of clarity, some details have been left out of (6.4.9). For example, in the update of Q_m , one would actually update only the first k columns (i.e. $Q_k \leftarrow$

$Q_m V_m(:, 1 : k))$, rather than updating the whole matrix and then discarding j columns.

Furthermore, the convergence test should be more comprehensive; it should test all of the subdiagonal entries $h_{i+1,i}$ for $i = 1, \dots, m-1$. If any one of them is effectively zero, then the first i columns of Q_k span an invariant subspace of dimension i . If $i < k$, then the iterations should continue, but the first i columns of Q_k should be locked in for the remainder of the computation. This saves some work in the update of Q_k .

Once some vectors have been locked in, one might choose to work with $m = k + j$ columns in addition to the i that have already been locked in, so that a total of $m + i$ columns are used during the next iteration. This is a good strategy in the sense that it promotes more rapid convergence. On the other hand, it uses more memory. If k eigenpairs are sought and we take $j = k$, then we need to be willing to store as many as $3k$ vectors.

One other convergence promoting strategy deserves mention. It is often helpful to work with spaces whose dimension is slightly greater than the number of eigenvalues wanted. For example, if we want s eigenvalues, we might work with spaces of dimension $k = s + 2$.

Nothing has been said in (6.4.9) about how the shifts should be chosen. As we have indicated above, they could be taken to be the j smallest eigenvalues of H_m . This is a good choice if the largest eigenvalues are sought. If, on the other hand, the eigenvalues of A that are furthest to the right in the complex plane are wanted, one can take the shifts to be the j leftmost eigenvalues of H_m , for example. Recall that we always want to choose shifts that are in the region of the spectrum that we are trying to suppress.

Carrying this idea further, if one is looking for eigenvalues that are close to some target τ (not on the periphery of the spectrum), one might take as shifts those eigenvalues of H_m that are furthest from τ . However, this strategy works poorly because it attempts to enhance interior eigenvalues, which the Arnoldi process does not produce well, while trying to suppress peripheral eigenvalues, which the the Arnoldi process is best at producing. A much more successful approach is to use the shift-and-invert strategy, that is, to apply IRA to the shifted, inverted operator $(A - \tau I)^{-1}$. This approach is feasible whenever we have the time and space to compute the LU decomposition of $(A - \tau I)$.

Why IRA Works

We stated above that an IRA iteration replaces the starting vector q by a new starting vector $\hat{q} = p(A)q$, where p is a polynomial chosen to suppress the unwanted eigenvectors and enhance the wanted ones. Now we will show that this is so. We begin with the Arnoldi configuration

$$AQ_m = Q_m H_m + q_{m+1} h_{m+1,m} e_m^T.$$

Clearly this equation continues to hold even after insertion of a shift:

$$(A - \nu_1 I)Q_m = Q_m(H_m - \nu_1 I) + E_1, \quad (6.4.10)$$

where the remainder term $E_1 = q_{m+1}h_{m+1,m}e_m^T$ is identically zero, except for the last column. Now consider applying a second shift. We can obtain an expression for $(A - \nu_2 I)(A - \nu_1 I)Q_m$ by multiplying (6.4.10) by $(A - \nu_2 I)$. The resulting equation contains an expression $(A - \nu_2 I)Q_m$, which can be eliminated by applying (6.4.10) with ν_1 replaced by ν_2 . The resulting expression has two remainder terms (terms involving E_1), which can be combined to yield

$$(A - \nu_2 I)(A - \nu_1 I)Q_m = Q_m(H_m - \nu_2 I)(H_m - \nu_1 I) + E_2,$$

where $E_2 = (A - \nu_2 I)E_1 + E_1(H_m - \nu_1 I)$ is identically zero, except in the last two columns. More generally, we have the following theorem, which is proved in Exercise 6.4.22.

Theorem 6.4.11 *Suppose*

$$AQ_m = Q_m H_m + q_{m+1}h_{m+1,m}e_m^T,$$

and let p be a polynomial of degree $j < m$. Then

$$p(A)Q_m = Q_m p(H_m) + E_j,$$

where $E_j \in \mathbb{C}^{n \times m}$ is identically zero, except in the last j columns.

Now let us recall what IRA does with Q_m and H_m . It chooses shifts $\nu_1, \dots, \nu_j$ from the region of the spectrum that is to be suppressed and does j steps of the QR algorithm on H_m using $\nu_1, \dots, \nu_j$ as shifts. This results in $\hat{H}_m = V_m^{-1}H_m V_m$, where V_m is the unitary factor in the QR decomposition $p(H_m) = V_m R_m$, as we recall from (6.4.3) and (6.4.4). Here $p(z) = (z - \nu_1) \cdots (z - \nu_j)$. The transforming matrix V_m is used to update Q_m to $\hat{Q}_m = Q_m V_m$. If we now apply Theorem 6.4.11, we obtain

$$p(A)Q_m = Q_m V_m R_m + E_j = \hat{Q}_m R_m + E_j. \quad (6.4.12)$$

Since we are interested in the first column, we multiply this equation by e_1 to get

$$p(A)Q_m e_1 = \hat{Q}_m R_m e_1 + E_j e_1.$$

The left-hand side of this equation is just $p(A)q_1$. On the right-hand side we have $R_m e_1 = \alpha e_1$, where $\alpha = r_{11} \neq 0$, because R_m is upper triangular. Therefore the first term on the right-hand side is just $\alpha \hat{q}_1$. The second term is zero, because the first column of E_j is zero. Thus we have

$$\hat{q}_1 = \beta p(A)q_1, \quad (6.4.13)$$

where $\beta = \alpha^{-1}$. This is exactly what we desired. Since $p(z)$ is zero at $\nu_1, \dots, \nu_j$, it follows that p will take on small values near these points and large values away

from these points. Referring back to (6.4.1), we see that eigenvectors corresponding to eigenvalues near $\nu_1, \dots, \nu_j$ will be suppressed, while those corresponding to eigenvalues away from $\nu_1, \dots, \nu_j$ will be enhanced.

IRA chooses different shifts on each iteration, but imagine for a moment a version that does not change the shifts. After i iterations the starting vector will be proportional to $p(A)^i q_1$, as we see by applying (6.4.13) i times. Thus IRA is just effecting the power method driven by $p(A)$. The eigenvectors that correspond to the largest eigenvalues $p(\lambda)$ of $p(A)$ will be favored.

The power method is subspace iteration with subspaces of dimension 1. Theorem 6.4.14 (below) shows that IRA also effects subspace iterations on higher dimensional spaces. Thus the convergence theory of subspace iteration can be brought to bear in the analysis of the convergence of IRA. In practice the subspace iterations are nonstationary, because different shifts are chosen at each step. This improves the algorithm by making it adaptive, hence flexible, but it makes the convergence analysis more difficult.

Theorem 6.4.14 *Suppose one iteration of IRA transforms $q_1, \dots, q_k$ to $\hat{q}_1, \dots, \hat{q}_k$. Let $p(z) = (z - \nu_1) \cdots (z - \nu_j)$, where $\nu_1, \dots, \nu_j$ are the shifts used by IRA. Then*

$$\text{span}\{\hat{q}_1, \dots, \hat{q}_i\} = p(A)\text{span}\{q_1, \dots, q_i\}, \quad \text{for } i = 1, \dots, k.$$

Proof. By Theorem 6.3.9, $\text{span}\{q_1, \dots, q_i\} = \mathcal{K}_i(A, q_1)$ and $\text{span}\{\hat{q}_1, \dots, \hat{q}_i\} = \mathcal{K}_i(A, \hat{q}_1)$. Furthermore (6.4.13) implies that $\mathcal{K}_i(A, \hat{q}_1) = p(A)\mathcal{K}_i(A, q_1)$, as one easily checks. $\square$

Exercise 6.4.15 Provide a second proof of Theorem 6.4.14 that uses (6.4.12), exploiting the special form of the error term E_j . $\square$

Algorithms of Davidson Type

The Arnoldi process builds up subspaces of increasing dimension by adding a vector at each step. This is done in such a way that at each step the space is a Krylov subspace: $\text{span}\{q_1, \dots, q_k\} = \mathcal{K}_k(A, q_1)$. Now we will consider methods that increase the subspace dimension by one at each step but do not use Krylov subspaces. A wide variety of algorithms results.

Suppose we have orthonormal vectors $q_1, \dots, q_k$, and we wish to add q_{k+1} . We want to pick this vector so that the expanded space is an improvement on the current space, in the sense that it contains better approximations to eigenvectors. For guidance we look to the space that we already have in hand. Let $Q_k = [q_1 \cdots q_k]$, as before, and let $B_k = Q_k^* A Q_k$. In the Arnoldi process this is just the Hessenberg matrix H_k , but now B_k need not be upper Hessenberg. Since k is not too big, we can compute its eigenvalues and eigenvectors easily. Let (μ, x) be some selected eigenpair (for example, take the largest μ), and let $q = Q_m x$. Then (μ, q) is a Ritz pair of A associated with the subspace $\text{span}\{q_1, \dots, q_k\}$ (see Exercise 6.3.35), and it can be viewed as an approximation to an eigenpair of A . The norm of the residual $r = Aq - \mu q$ gives an indication of the quality of (μ, q) as an approximate eigenpair.

Several important methods make use of the residual to determine the new subspace vector q_{k+1} . In each case r is used to determine a second vector s such that $s \notin \text{span}\{q_1, \dots, q_k\}$. Then q_{k+1} is obtained by orthonormalizing s against $q_1, \dots, q_k$ by the Gram-Schmidt process. The methods differ in how they obtain s from r .

The simplest possibility is to take $s = r$. This leads to a method that is equivalent to the Arnoldi process. See Exercise 6.4.23.

A second possibility is to take $s = (D - \mu I)^{-1}r$, where D is the diagonal matrix that has the same main-diagonal entries as A . This leads to *Davidson's method*, which has been used extensively in quantum chemistry calculations. In these applications the matrices are symmetric and extremely large. They are also strongly diagonally dominant, which means that the main-diagonal entries are much larger than the entries off the main diagonal. This property is crucial to the success of Davidson's method. Notice that the computation of s is quite inexpensive, since $(D - \mu I)^{-1}$ is a diagonal matrix.

A third way of choosing s leads to the *Jacobi-Davidson* method, which we motivate as follows. If q is close to an eigenvector of A , then a small correction $\tilde{s}$ can make $q + \tilde{s}$ an exact eigenvector. Thus

$$A(q + \tilde{s}) = (\mu + \tilde{\nu})(q + \tilde{s}), \quad (6.4.16)$$

where $\tilde{\nu}$ is a small correction to the Ritz value μ . Furthermore, we may take the correction to be orthogonal to q , i.e. $q^* \tilde{s} = 0$. The Jacobi-Davidson method chooses s to be an approximation to $\tilde{s}$. Suppose $\|\tilde{s}\| = O(\epsilon)$ and $|\tilde{\nu}| = O(\epsilon)$, where $\epsilon \ll 1$. Expanding (6.4.16) we obtain $Aq + A\tilde{s} = \mu q + \tilde{\nu}q + \mu\tilde{s} + \tilde{\nu}\tilde{s}$, or

$$(A - \mu I)\tilde{s} - q\tilde{\nu} = -r + \tilde{\nu}\tilde{s}.$$

We expect the term $\tilde{\nu}\tilde{s}$ to have a negligible effect on this equation, because $\|\tilde{\nu}\tilde{s}\| = O(\epsilon^2)$. The Jacobi-Davidson method takes s and ν to be the solution to the equation obtained by ignoring this term while enforcing the orthogonality condition $q^*s = 0$. Thus

$$(A - \mu I)s - q\nu = -r, \quad q^*s = 0,$$

or

$$\begin{bmatrix} A - \mu I & q \\ q^* & 0 \end{bmatrix} \begin{bmatrix} s \\ -\nu \end{bmatrix} = \begin{bmatrix} -r \\ 0 \end{bmatrix}. \quad (6.4.17)$$

The reader might recognize this as Newton's method. It is also a variant of Rayleigh quotient iteration (Exercise 6.4.24).

To summarize, the Jacobi-Davidson method obtains a Ritz pair (μ, q) from $\text{span}\{q_1, \dots, q_k\}$, computes the residual $r = Aq - \mu q$, then solves (6.4.17) to obtain s (and ν). The new vector q_{k+1} is obtained by orthogonalizing s against $q_1, \dots, q_k$ by some version of Gram Schmidt. The new space $\text{span}\{q_1, \dots, q_{k+1}\}$ contains both q and s , so it also contains $q + s$, an excellent approximation to an eigenvector.

What we have just described is known as the "exact" variant of Jacobi-Davidson, since the equation (6.4.17) is solved exactly. In practice, if A is extremely large, it will

be impractical to solve (6.4.17). Therefore the Jacobi-Davidson process is usually used in an “inexact” mode, which is to say that (6.4.17) is solved approximately by an iterative method such as those discussed in Section 7.9. An accurate solution is not needed; a very crude approximation is often good enough. The only penalty for using a poor approximation is that the vector that is added to the space is of reduced quality. This means that more steps will need to be taken before a sufficiently accurate approximate eigenpair is obtained. We do not mind taking many steps if the cost of each step is not too great. There are many variations on this theme and a number of ways of rewriting the system of equations (6.4.17). For more information see [3] or the award-winning paper [62].

At each step of the Jacobi-Davidson procedure, we have k Ritz pairs to choose from, where k is the current dimension of the space. Which pair we actually choose for our next step depends upon our objective. If we wish to find the largest eigenvalues of A , we should choose the largest Ritz value. If, on the other hand, we wish to find interior eigenvalues near the target value τ , it makes sense to choose the Ritz value that is closest to τ . However, in this context it is usually better to work with *harmonic* Ritz values instead of the standard Ritz values. See Exercise 6.4.25.

Each step of the Davidson or Jacobi-Davidson algorithm adds a new vector. After many steps we may wish to dispose of the less promising vectors and keep only a small subspace that contains the best estimates of eigenvectors, as is done in IRA. This is easier in the current context than it is in IRA; the subspaces that we are building are not Krylov subspaces, so we do not need to worry about preserving that property. We can proceed as follows, for example. Suppose we have $m = k + j$ orthonormal vectors, the columns of the matrix $Q \in \mathbb{C}^{n \times m}$, and we want to discard j columns and keep a k -dimensional subspace. Let $B = Q^*AQ$. Compute the Schur decomposition $B = UTU^*$. U is unitary, and T is upper triangular. The main diagonal entries of T are the eigenvalues of B , which are Ritz values of A with respect to the current subspace. These can be made to appear in any order in T . Suppose we order them so that

$$T = \begin{bmatrix} T_{11} & T_{12} \\ 0 & T_{22} \end{bmatrix},$$

where $T_{11} \in \mathbb{C}^{k \times k}$ contains the k “most promising” Ritz values, the ones that we want to keep. Let $\hat{Q} = QU$, and partition $\hat{Q}$ conformably with T , i.e. $\hat{Q} = \begin{bmatrix} \hat{Q}_1 & \hat{Q}_2 \end{bmatrix}$, where $\hat{Q}_1 \in \mathbb{C}^{n \times k}$. Then $T_{11} = \hat{Q}_1^*A\hat{Q}_1$, which implies that the eigenvalues of T_{11} are the Ritz values of A with respect to the space $\mathcal{R}(\hat{Q}_1)$. If we now keep $\hat{Q}_1$ and discard $\hat{Q}_2$, we will have retained the desired Ritz values and a space that contains their associated Ritz vectors. This process is known as *purging*.

Exercise 6.4.18 Check the claims of the previous paragraph. In particular, show that $T_{11} = \hat{Q}_1^*A\hat{Q}_1$. $\square$

In situations where more than one eigenvalue is sought, it will often happen that some eigenvalues converge before the others do. Then we have an invariant subspace that we wish to preserve while looking for a larger invariant subspace. It is a simple

matter to lock in the converged space. We simply move the converged vectors to the front of our basis (if they are not already there) and keep them there.

Additional Exercises

Exercise 6.4.19 MATLAB's built-in `eigs` function computes a few eigenvalues of a sparse (or full) matrix by the implicitly-restarted Arnoldi (IRA) method. In MATLAB type `help eigs` to learn how the `eigs` command works. Use `eigs` to calculate the ten eigenvalues of `west0479` closest to $\tau = 6 + 0i$. Use a residual tolerance of 10^{-16} . How many iterations did `eigs` take? Did this method take more or less work than straight shift-and-invert Arnoldi? $\square$

Exercise 6.4.20 Let $v_1, \dots, v_k$ be eigenvectors of A , and suppose $q = c_1 v_1 + \dots + c_k v_k$.

- Show that for $j = 1, 2, 3, \dots$, we have $A^j q \in \text{span}\{v_1, \dots, v_k\}$. Deduce that the Krylov subspaces $\mathcal{K}_m(A, q)$ lie in $\text{span}\{v_1, \dots, v_k\}$ for all m .
- Show that $\mathcal{K}_k(A, q)$ is invariant under A . Deduce that the Arnoldi process starting with q will terminate in k steps or fewer with $\text{span}\{q_1, \dots, q_m\}$ ($m \leq k$) an invariant subspace contained in $\text{span}\{v_1, \dots, v_k\}$.
- Let $\lambda_1, \dots, \lambda_k$ be the eigenvalues of A associated with $v_1, \dots, v_k$, respectively. Show that if the λ_j are all distinct and $c_j \neq 0$ for all j , then $q, Aq, A^2q, \dots, A^{k-1}q$ are linearly independent, and the Arnoldi process terminates in exactly k steps with $\text{span}\{q_1, \dots, q_k\} = \mathcal{K}_k(A, q) = \text{span}\{v_1, \dots, v_k\}$.
- Suppose some of the eigenvalues $\lambda_1, \dots, \lambda_k$ are equal. Show that if there are only s distinct eigenvalues, $s < k$, among $\lambda_1, \dots, \lambda_k$, then q can be written as a linear combination of s or fewer eigenvectors, and the Arnoldi process will terminate in s steps or fewer.

$\square$

Exercise 6.4.21 Let j be a non-negative integer. A matrix B is called j -Hessenberg if $b_{ik} = 0$ whenever $i - k > j$. A j -Hessenberg matrix is said to be *properly* j -Hessenberg if $b_{ij} \neq 0$ whenever $i - k = j$.

- What are the common names for 0-Hessenberg and 1-Hessenberg matrices? Give two examples of 2-Hessenberg matrices of dimension 5, one properly 2-Hessenberg and one not.
- Show that the product of a properly j -Hessenberg matrix and a properly k -Hessenberg matrix is a properly $(j + k)$ -Hessenberg matrix.
- Show that if $B \in \mathbb{C}^{m \times m}$ is properly j -Hessenberg ($j < m$), then the first $m - j$ columns of B are linearly independent.
- Show that if H_m is properly upper Hessenberg and p is a polynomial of degree j , say $p(z) = (z - \nu_1)(z - \nu_2) \cdots (z - \nu_j)$, then $p(H_m)$ is properly j -Hessenberg.

(e) Partition the QR decomposition $p(H_m) = V_m R_m$ of (6.4.4) as follows:

$$\begin{bmatrix} B_1 & B_2 \end{bmatrix} = \begin{bmatrix} V_1 & V_2 \end{bmatrix} \begin{bmatrix} R_{11} & R_{12} \\ 0 & R_{22} \end{bmatrix},$$

where $B_1 \in \mathbb{C}^{m \times j}$, $V_1 \in \mathbb{C}^{m \times j}$, and $R_{11} \in \mathbb{C}^{j \times j}$. From parts (c) and (d), deduce that the columns of B_1 are linearly independent. Show that this implies that R_{11} is nonsingular, and $V_1 = B_1 R_{11}^{-1}$.

(f) Deduce from part (e) that V_m is a properly j -Hessenberg matrix. Conclude that the first $m - j - 1$ entries of the row vector $e_m^T V_m$ are zero, and the $m - j$ th entry (which we called β in (6.4.6)) is nonzero.

□

Exercise 6.4.22 Prove Theorem 6.4.11 by induction on j :

- (a) Show that the theorem holds when $j = 1$. In this case $p(z) = \alpha_1(z - \nu_1)$, where α_1 is a nonzero constant.
- (b) Show that the theorem holds when $j = 2$. In this case $p(z) = \alpha_2(z - \nu_1)(z - \nu_2)$. This step is just for practice; it is not crucial to the proof of the Theorem.
- (c) Show that if the theorem holds for polynomials of degree $j - 1$, then it holds for polynomials of degree j .

□

Exercise 6.4.23 Let v be a Ritz vector from $\text{span}\{q_1, \dots, q_k\}$. Then $v = c_1 q_1 + \dots + c_k q_k$ for some constants $c_1, \dots, c_k$. Typically $c_k \neq 0$. Consider a method that expands the subspace at each step by picking a Ritz pair (μ, v) at each step, computing the residual $r = Av - \mu v$, and orthonormalizing the residual against the previously determined vectors. Using induction on k , show that if $c_k \neq 0$ at each step, then $\text{span}\{q_1, \dots, q_k\} = \mathcal{K}_k(A, q_1)$ for all k . Thus this method is equivalent to the Arnoldi process. What happens if $c_k = 0$ at some step?

□

Exercise 6.4.24 Let (μ, q) be a Ritz pair of A from some subspace. Show that $\mu = q^* A q / q^* q$. Thus μ is the Rayleigh quotient of q . Show that if s is determined by the Jacobi-Davidson equation (6.4.17), then $q + s = \nu(A - \mu I)^{-1} q$. Therefore $q + s$ is the result of one step of Rayleigh quotient iteration with starting vector q .

□

Exercise 6.4.25 In this exercise we introduce *harmonic* Ritz pairs. We use the same notation as in Exercise 6.3.35. Let τ be a complex (target) value that is not an eigenvalue of A . Then μ is a *harmonic Ritz value* of A with target τ with respect to the space $\mathcal{S}$ if $(\mu - \tau)^{-1}$ is an ordinary Ritz value of $(A - \tau I)^{-1}$. From Exercise 6.3.35 we see that μ satisfies this property if and only if $(\mu - \tau)^{-1}$ is an eigenvalue of $Q^*(A - \tau I)^{-1}Q$. Although this is a small ($k \times k$) matrix, it can be difficult to evaluate if A is large. It turns out to be easier to obtain Harmonic Ritz values with respect to a different space. Let $\mathcal{U} = (A - \tau I)\mathcal{S}$.

- (a) Let $u \in \mathbb{C}^n$. Show that $u \in \mathcal{U}$ if and only if there is an $x \in \mathbb{C}^k$ such that $u = (A - \tau I)Qx$. Show that the vector x corresponding to a fixed $u \in \mathcal{U}$ is uniquely determined.
- (b) Show that μ is a harmonic Ritz value of A with target τ with respect to the space $\mathcal{U}$ if and only if there is a $u \in \mathcal{U}$ such that

$$(A - \tau I)^{-1}u - (\mu - \tau)^{-1}u \perp \mathcal{U},$$

and that this holds in turn if and only if there is an $x \in \mathbb{C}^k$ such that

$$(A - \mu I)Qx \perp \mathcal{U}.$$

Let $v = Qx$. We call v a *harmonic Ritz vector* associated with the harmonic Ritz value μ . Notice that v belongs to $\mathcal{S}$, not $\mathcal{U}$.

- (c) Let $Y = (A - \tau I)Q \in \mathbb{C}^{n \times k}$. Show that μ is a harmonic Ritz value (with τ and $\mathcal{U}$), with associated harmonic Ritz vector $v = Qx$ if and only if

$$Y^*Yx = (\mu - \tau)Y^*Qx.$$

This equation is an example of a *generalized* eigenvalue problem (see Section 6.7). Since the matrices Y^*Y and Y^*Q are small ($k \times k$), this problem can be solved inexpensively.

- (d) Show that μ is a harmonic Ritz value (with τ and $\mathcal{U}$) if and only if $\mu - \tau$ is an eigenvalue of $(Y^*Y)(Y^*Q)^{-1}$ and $(Y^*Q)^{-1}(Y^*Y)$, assuming that $(Y^*Q)^{-1}$ exists.

□

6.5 SENSITIVITY OF EIGENVALUES AND EIGENVECTORS

Since the matrix A whose eigensystem we wish to calculate is never known exactly, it is important to study how the eigenvalues and eigenvectors are affected by perturbations of A . Thus in this section we will ask the question of how close the eigenvalues and eigenvectors of $A + \delta A$ are to those of A if $\|\delta A\|/\|A\|$ is small. This would be an important question even if the uncertainty in A were our only concern, but of course there is also another reason for asking it. We noted in Chapter 3 that any algorithm that transforms a matrix by rotators or reflectors constructed as prescribed in Section 3.2 is backward stable. The implicit QR algorithms are of this type. This means that an implicit QR algorithm determines the exact eigenvalues of a matrix $A + \delta A$ where $\|\delta A\|/\|A\|$ is small. If we can show that the eigenvalues of $A + \delta A$ are close to those of A , we will know that our answers are accurate. Of course it will turn out that we cannot always guarantee accurate eigenvalues; the accuracy depends upon certain condition numbers.

A related question is that of residuals. Suppose we have calculated an approximate eigenvalue λ and associated eigenvector v , and we wish to know whether they are accurate. It is natural to calculate the residual $r = Av - \lambda v$ and check whether it is small. Suppose $\|r\|$ is small. Does this guarantee that λ and v are accurate? As the following theorem shows, this question also reduces to that of the sensitivity of A .

Theorem 6.5.1 *Let $A \in \mathbb{C}^{n \times n}$, let v be an approximate eigenvector of A with $\|v\|_2 = 1$, λ an associated approximate eigenvalue, and r the residual: $r = Av - \lambda v$. Then λ and v are an exact eigenpair of some perturbed matrix $A + \delta A$, where $\|\delta A\|_2 = \|r\|_2$.*

Proof. If you have worked Exercise 6.3.36, then you have already done this. Let $\delta A = -rv^*$. Then $\|\delta A\|_2 = \|r\|_2 \|v\|_2 = \|r\|_2$, and $(A + \delta A)v = Av - rv^*v = Av - r = \lambda v$. $\square$

Exercise 6.5.2 Verify that $\|rv^*\|_2 = \|r\|_2 \|v\|_2$ for all $r, v \in \mathbb{C}^n$. (On the left-hand side of this equation we have the matrix 2-norm; on the right-hand side we have the vector 2-norm, the Euclidean norm.) $\square$

Sensitivity of Eigenvalues

First of all, the eigenvalues of a matrix depend continuously on the entries of the matrix. This is so because the coefficients of the characteristic polynomial are continuous functions of the matrix entries, and the zeros of the characteristic polynomial, that is, the eigenvalues, depend continuously on the coefficients. See [81] and the references cited therein.² This means that we can put as small a bound as we please on how far the eigenvalues can wander, simply by making the perturbation of the matrix sufficiently small. But this information is too vague. It would be more useful to have a number κ such that if we perturb the matrix by ϵ , then the eigenvalues are perturbed by at most $\kappa\epsilon$, at least for sufficiently small ϵ . Then κ would serve as a condition number for the eigenvalues. It turns out that we can get such a κ if the matrix is semisimple. This is the content of our next theorem, which is due to Bauer and Fike.

Theorem 6.5.3 *Let $A \in \mathbb{C}^{n \times n}$ be a semisimple matrix, and suppose $V^{-1}AV = D$, where V is nonsingular and D is diagonal. Let $\delta A \in \mathbb{C}^{n \times n}$ be some perturbation of A , and let μ be an eigenvalue of $A + \delta A$. Then A has an eigenvalue λ such that*

$$|\mu - \lambda| \leq \kappa_p(V) \|\delta A\|_p \quad (6.5.4)$$

for $1 \leq p \leq \infty$.

This theorem shows that $\kappa_p(V)$ (which was defined in Section 2.2) is an overall condition number for the spectrum of A . We could nondimensionalize (6.5.4) by dividing both sides by $\|A\|_p$, but that would be an unnecessary complication.

²Two other approaches are outlined in [15] and [73].

Proof. Let $\delta D = V^{-1}(\delta A)V$. Then

$$\|\delta D\|_p \leq \|V^{-1}\|_p \|\delta A\|_p \|V\|_p = \kappa_p(V) \|\delta A\|_p. \quad (6.5.5)$$

Since $D + \delta D$ is similar to $A + \delta A$, μ is an eigenvalue of $D + \delta D$. Let x be an associated eigenvector. If μ happens to be an eigenvalue of A , we are done, so suppose it is not. Then $\mu I - D$ is nonsingular, and the equation $(D + \delta D)x = \mu x$ can be rewritten as $x = (\mu I - D)^{-1}(\delta D)x$. Thus $\|x\|_p \leq \|(\mu I - D)^{-1}\|_p \|\delta D\|_p \|x\|_p$. Canceling out the factor $\|x\|_p$ and rearranging, we find that

$$\|(\mu I - D)^{-1}\|_p^{-1} \leq \|\delta D\|_p. \quad (6.5.6)$$

The matrix $(\mu I - D)^{-1}$ is diagonal with main diagonal entries $(\mu - \lambda_i)^{-1}$, where $\lambda_1, \dots, \lambda_n$ are the eigenvalues of A . It follows very easily that $\|(\mu I - D)^{-1}\|_2 = |\mu - \lambda|^{-1}$, where λ is the eigenvalue of A that is closest to μ . Thus (6.5.6) can be rewritten as

$$|\mu - \lambda| \leq \|\delta D\|_p.$$

Combining this inequality with (6.5.5), we get our result. $\square$

Exercise 6.5.7 Verify the following facts, which were used in the proof of Theorem 6.5.3

- (a) Let Δ be a diagonal matrix with main diagonal entries $\delta_1, \dots, \delta_n$. Verify that for $1 \leq p \leq \infty$, $\|\Delta\|_p = \max_{1 \leq i \leq n} |\delta_i|$.
- (b) Verify that $\|(\mu I - D)^{-1}\|_p = |\mu - \lambda|^{-1}$, where λ is the eigenvalue of the diagonal matrix D that is closest to μ .

$\square$

The columns of the transforming matrix V are eigenvectors of A , so the condition number $\kappa_p(V)$ is a measure of how far from being linearly dependent the eigenvectors are: the larger the condition number, the closer they are to being dependent. From this viewpoint it would seem reasonable to assign an overall condition number of infinity to the eigenvalues of a defective matrix, since such matrices do not even have n linearly independent eigenvectors. Support for this viewpoint is given in Exercise 6.5.32.

If A is Hermitian or even normal, V can be taken to be unitary (cf. Theorems 5.4.12 and 5.4.15). Unitary V satisfy $\kappa_2(V) = 1$, so the following corollary holds.

Corollary 6.5.8 Let $A \in \mathbb{C}^{n \times n}$ be normal, let δA be any perturbation of A , and let μ be any eigenvalue of $A + \delta A$. Then A has an eigenvalue λ such that

$$|\mu - \lambda| \leq \|\delta A\|_2.$$

Corollary 6.5.8 can be summarized by saying that the eigenvalues of a normal matrix are perfectly conditioned. If a normal matrix is perturbed slightly, the resulting

perturbation of the eigenvalues is no greater than the perturbation of the matrix elements.

A weakness of Theorem 6.5.3 is that it gives a single overall condition number for the eigenvalues. In fact it can happen that some of the eigenvalues are well conditioned while others are ill conditioned. This is true for both semisimple and defective matrices. It is therefore important to develop individual condition numbers for the eigenvalues. Again we will restrict our attention to the semisimple case; in fact, we will assume distinct eigenvalues.

Our discussion of individual condition numbers will depend on the notion of *left eigenvectors*. Let λ be an eigenvalue of A . Then $A - \lambda I$ is singular, so there exists a nonzero $x \in \mathbb{C}^n$ such that $(A - \lambda I)x = 0$, that is,

$$Ax = \lambda x. \quad (6.5.9)$$

It is equally true that there is a nonzero $y \in \mathbb{C}^n$ such that $y^*(A - \lambda I) = 0$, that is,

$$y^*A = \lambda y^*. \quad (6.5.10)$$

Any nonzero vector y^* that satisfies (6.5.10) is called a *left eigenvector* of A associated with the eigenvalue λ . Actually we will be somewhat casual with the nomenclature and refer to y itself as a left eigenvector of A , as if y and y^* were the same thing. Furthermore, to avoid confusion we will refer to any nonzero x satisfying (6.5.9) as a *right eigenvector* of A . A left eigenvector of A is just (the conjugate transpose of) a right eigenvector of A^* .

Theorem 6.5.11 *Let $A \in \mathbb{C}^{n \times n}$ have distinct eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$ with associated linearly independent right eigenvectors $x_1, \dots, x_n$ and left eigenvectors $y_1, \dots, y_n$. Then*

$$y_j^* x_i \quad \begin{cases} = 0 & \text{if } i \neq j \\ \neq 0 & \text{if } i = j \end{cases}$$

(Two sequences of vectors that satisfy these relationships are said to be biorthogonal.)

Proof. Suppose $i \neq j$. From the equation $Ax_i = \lambda_i x_i$ it follows that $y_j^* Ax_i = \lambda_i y_j^* x_i$. On the other hand, $y_j^* A = \lambda_j y_j^*$, so $y_j^* Ax_i = \lambda_j y_j^* x_i$. Thus $\lambda_i y_j^* x_i = \lambda_j y_j^* x_i$. Since $\lambda_i \neq \lambda_j$, it must be true that $y_j^* x_i = 0$.

It remains to be shown that $y_i^* x_i \neq 0$. Let's assume that $y_i^* x_i = 0$ and get a contradiction. From our assumption and the first part of the proof we see that $y_i^* x_k = 0$ for $k = 1, \dots, n$. The vectors $x_1, \dots, x_n$ are linearly independent, so they form a basis for $\mathbb{C}^n$. Notice that $y_i^* x_k$ is just the complex inner product (x_k, y_i) , so y_i is orthogonal to $x_1, \dots, x_n$, hence to every vector in $\mathbb{C}^n$. In particular it is orthogonal to itself, which implies $y_i = 0$, a contradiction. $\square$

The next theorem establishes the promised condition numbers for individual eigenvalues. Let's take a moment to set the scene. Suppose $A \in \mathbb{C}^{n \times n}$ has n distinct eigenvalues, and let λ be one of them. Let δA be a small perturbation satisfying $\|\delta A\|_2 = \epsilon$. Since the eigenvalues of A are distinct, and they depend continuously